

Project Data Page:

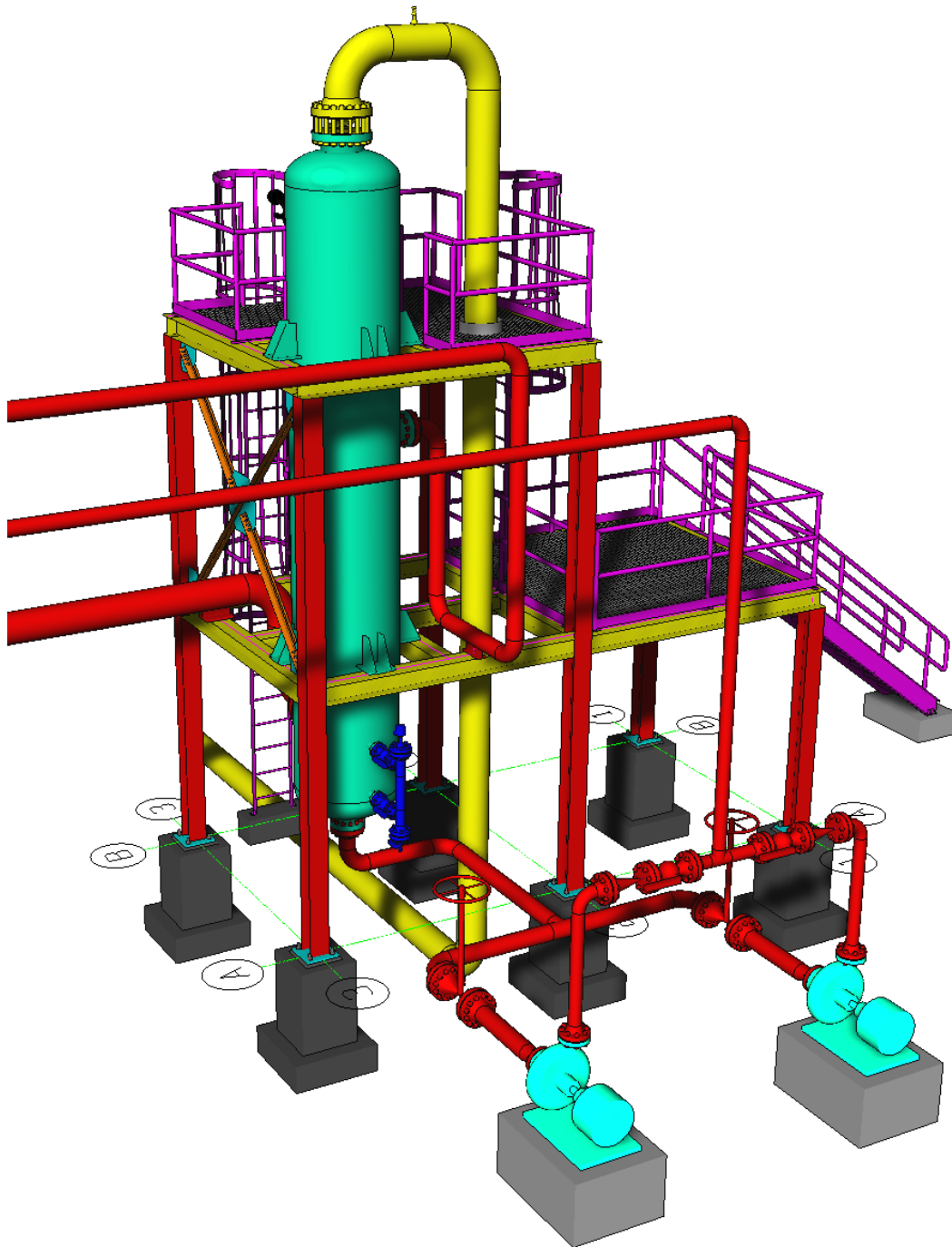




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\408-V-005AB0operating7%.pvdb

Date of Analysis : May 19, 2026 4:17pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 3500.000 mm. ID
 Vessel Design Length, Tangent to Tangent 10100.00 mm.
 Distance of Bottom Tangent above Grade 2300.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 2050.00 mm.
 Internal Design Temperature 65 °C
 Internal Design Pressure 13.000 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 17 bars
 Shop Test Pressure 1.000 bars
 Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature 4.0 °C
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code Response Spectrum

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom D end	...	...	...	1.5000	N/A	No
Bottom D end	13.001	0.00	18.000	3.0000	No	No
Shell 1	13.001	0.00	17.000	3.0000	N/A	No
Shell 2	13.001	0.00	17.000	3.0000	N/A	No
Shell 2	13.001	0.00	17.000	3.0000	N/A	No
Shell 3	13.000	0.00	17.000	3.0000	N/A	No
Top D end	13.000	0.00	18.000	3.0000	No	No

Liquid Level: 11850.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.001

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-50.0	2000.0	8.0	8.0	...	...	1.00	1.00



Ellipse	0.0	50.0	24.0	20.0	15.5	...	1.00	1.00
Cylinder	3000.0	3000.0	24.0	24.0	19.4	...	0.85	0.85
Cylinder	6000.0	3000.0	24.0	24.0	19.4	...	0.85	0.85
Cylinder	8000.0	2000.0	24.0	24.0	19.4	...	0.85	0.85
Cylinder	10000.0	2000.0	24.0	24.0	19.4	...	0.85	0.85
Ellipse	10050.0	50.0	24.0	20.0	15.5	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_3"	WRC-107/537	0.434	n/a	Passed
N1_4"	WRC-297	0.848	Shell	Passed
N4_3"	WRC-107/537	0.434	n/a	Passed
N2_4"	WRC-107/537	0.434	n/a	Passed
N5_3"	WRC-107/537	0.434	n/a	Passed
N6_3"	WRC-107/537	0.434	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kg	Earthquake Shear Kg	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	12906.1	9965.06	104063	106271
20	30	2025	11706.1	9905.11	79450.5	86431.1
30	40	3550	11676	9726.36	78865.9	85938.1
40	50	6550	9367.68	9053.29	47299.8	56840.6
50	60	9050	6802.37	8021.22	23044.2	29707.8
60	70	11050	5593.85	6758.55	10647.8	13668.7
70	80	12075	3602.16	3026.87	1451.57	151.347

Abs Max of the all of the Stress Ratio's : 0.6873

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	25.0000	mm.
Inside Diameter of Basing	3430.0000	mm.
Outside Diameter of Basing	3760.0000	mm.
Nominal Diameter of Bolts	30.0000	mm.
Diameter of Bolt Circle	3639.9998	mm.
Number of Bolts	20	

Thickness of Gusset Plates	10.0000	mm.
Average Width of Gusset Plates	102.0000	mm.
Height of Gussets	185.0000	mm.
Distance between Gussets	90.0000	mm.
Thickness of Top Plate or Ring	25.0000	mm.
Radial Width of the Top Plate	120.0000	mm.

Loads for Foundation/Support Design:

Total Wind Shear on Support	16078.	Kgf
Total Earthquake Shear on Support	13137.	Kgf
Wind Moment on Support	201623.	Kg-m.
Earthquake Moment on Support	203831.	Kg-m.
Maximum Initial Bolt Load	7593.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	34723.4	kg.
Shop Test - Fabricated + water (full)	154074.7	kg.
Shipping - Fab. + rem. intls.+ shipping app.	34723.4	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	70472.3	kg.

PV Elite 28 Licensee: SL User
FileName : 408-V-005AB_Operating_7%
Vessel Design Summary:

Step: 39 4:17pm May 19,2026

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HEXAGON

Empty	- Fab. + intls. + details + wghts.	70472.3	kg.
Operating	- Empty + operating liquid (no ca)	70602.3	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
N3_3"	18	...	...	OK	OK	Passed
MH1_24"	19	...	...	...	OK	Passed
N1_4"	19	...	...	OK	OK	Passed
N4_3"	18	...	...	OK	OK	Passed
N2_4"	18	...	...	OK	OK	Passed
N5_3"	18	...	...	OK	OK	Passed
N6_3"	18	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 18 Nozzle : N33"
 Minimum MAWP Shells/Flanges : 17 Element : Shell 1
 Minimum MAPnc Shells/Flanges : 20 Element : Shell 1

Computed Vessel M.A.W.P. : 17 bars

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Inside Radius mm.
20	N3_3"	0.000	0.000	37.715
30	MH1_24"	1050.000	0.000	292.531
30	N1_4"	400.000	180.000	49.025
70	N4_3"	0.000	0.000	37.715
70	N2_4"	0.000	0.000	50.415
70	N5_3"	0.000	90.000	37.715
70	N6_3"	0.000	180.000	37.715

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3_3"	3.000 in	160	WNF	3.500	11.125	178.90	20.00	220.31	150
N4_3"	3.000 in	160	WNF	3.500	11.125	166.20	20.00	420.31	150
N5_3"	3.000 in	160	WNF	3.500	11.125	166.20	20.00	436.24	300
N6_3"	3.000 in	160	WNF	3.500	11.125	166.20	20.00	436.24	300
N1_4"	4.000 in	120	WNF	4.500	11.125	204.30	24.00	224.93	300
N2_4"	4.000 in	120	WNF	4.500	11.125	204.30	20.00	440.70	300
MH1_24"	24.000 in	40	WNF	24.000	17.450	1119.60	24.00	300.75	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3_3"	SA-106 B	20.000	20.000	20.000	20.000	...
N4_3"	SA-106 B	20.000	20.000	20.000	20.000	...
N5_3"	SA-106 B	20.000	20.000	20.000	20.000	...
N6_3"	SA-106 B	20.000	20.000	20.000	20.000	...
N1_4"	SA-106 B	20.000	20.000	20.000	24.000	...
N2_4"	SA-106 B	20.000	20.000	20.000	20.000	...
MH1_24"	SA-516 70	20.000	20.000	20.000	24.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N3_3"	...	0.0	200.00	0.00	Bottom D end
N4_3"	...	0.0	400.00	0.00	Top D end
N5_3"	...	90.0	400.00	0.00	Top D end
N6_3"	...	180.0	400.00	0.00	Top D end
N1_4"	350.000	180.0	200.00	0.00	Shell 1
N2_4"	...	0.0	400.00	0.00	Top D end
MH1_24"	1000.000	0.0	250.00	0.00	Shell 1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 24.0mm. THK X 3500.0mm. ID X 50.0mm.	SA-516 70
3	PLATFORM: 1200mm X 1500mm WIDE	
1	PLATFORM: 1200mm X 9550mm WIDE	
1	LADDER: 11000.0mm	...
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	CLASS 150 GR 1.1, 3.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 3.0" BLIND FLANGE(S)	SA-105
1	BASERING: 3760.0mm. OD X 3430.0 ID X 25.0mm.	SA-516 70
20	BASERING BOLTS: 30.0mm. DIA	SA-36
20	BASERING BOLT NUTS: 30.0mm. DIA	...
1	BASERING TOP RING: 3770.0mm. OD X 3530.0 ID X 25.0mm.	...
40	BASERING GUSSET PLATES: 102.0mm. X 102.0mm. X 185.0mm. X 10.0mm.	...
3	CON SKIRT SEGMENT 8.0mm. THK X 3060.1mm. X 2000.0mm. X 3048.0mm.	SA-516 70
1	CON SKIRT SEGMENT 8.0mm. THK X 1858.9mm. X 2000.0mm. X 1851.5mm.	SA-516 70
2	CYLINDRICAL SEGMENT 24.0mm. THK X 5497.7mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 24.0mm. THK X 3048.0mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm. THK X 7947.4mm. X 3000.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 24.0mm. THK X 5497.7mm. X 2000.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
Notes							
Bottom D end [10]	B	9	4	No	0.882	20.000	1.00
Bottom D end [7]	B	13	4	No	0.796	24.000	1.00
Shell 1 [8]	B	13	4	No	0.796	24.000	0.85
Shell 2 [8]	B	-5	-16	Yes	0.796	24.000	0.85
Shell 2 [8]	B	13	4	No	0.796	24.000	0.85
Shell 3 [8]	B	-5	-16	Yes	0.796	24.000	0.85
Top D end [10]	B	9	4	No	0.882	20.000	1.00
Top D end [7]	B	13	4	No	0.796	24.000	1.00
N33" [1]	B	9	4	No	0.882	20.000	1.00
Nozzle Flg [5]	A	-18	-26	No			
MH124" [1]	B	13	4	No	0.796	24.000	1.00
Nozzle Flg [5]	B	-46	-48	No			
N14" [1]	B	13	4	No	0.796	24.000	1.00
Nozzle Flg [5]	B	-46	-104	No			
N43" [1]	B	9	4	No	0.882	20.000	1.00
Nozzle Flg [5]	A	-18	-26	No			
N24" [1]	B	9	4	No	0.882	20.000	1.00
Nozzle Flg [5]	A	-18	-104	No			
N53" [1]	B	9	4	No	0.882	20.000	1.00
Nozzle Flg [5]	A	-18	-104	No			
N63" [1]	B	9	4	No	0.882	20.000	1.00
Nozzle Flg [5]	A	-18	-104	No			
Nozl. Bolting[22]		-48		No	1.000	64.000	1.00

Warmest MDMT: 13 4

Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature 4.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Warn. 10 20 Skirt diameter at base is < at the top

Class From To: Check of Additional Element Data

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest) 13 bars
 Design Internal Temperature 65.0 °C
 Projection of Nozzle from Vessel Top 0 mm.
 Projection of Nozzle from Vessel Bottom 0 mm.
 Minimum Design Metal Temperature 11.4 °C
 Type of Construction Welded
 Special Service Amine
 Use Higher Longitudinal Stresses (Flag) N
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test User Defined Hydro
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE
 Load Case 13 IP+WE+EW
 Load Case 14 IP+WF+CW
 Load Case 15 IP+VO+OW
 Load Case 16 IP+VE+EW
 Load Case 17 NP+VO+OW
 Load Case 18 FS+BS+IP+OW
 Load Case 19 FS+BS+EP+OW
 Load Case 20 BL+IP+OW



Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Bottom D end	
Distance "FROM" to "TO"	2000	mm.
Skirt Outside Diameter	3530	mm.
Diameter of Skirt at Base	3516	mm.
Skirt Thickness	8	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	8	mm.
External Corrosion Allowance	0	mm.
Design Temperature Internal Pressure	65	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	175	N./mm2
Allowable Stress, Operating	164	N./mm2
Allowable Stress, Hydrotest	247	N./mm2
Material Density	0.00775	kg./cm3
P Number Thickness	0	mm.
Yield Stress, Operating	246	N./mm2
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Head-to-Skirt or Circ. Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	skirt manway wt	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	330	Kgf
Offset from Element Centerline	0	mm.

Element#	2/7	
Element From Node	20	
Element To Node	30	
Element Type	Elliptical	
Description	Bottom D end	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	3500	mm.
Element Thickness	20	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	24	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	-875	mm.
Height/Length of Liquid	925	mm.
Liquid Density	0.11981E-05	kg./cm3
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N33"	
Dist. from "FROM" Node / Offset dist	0	mm.



Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	51.36	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell 1
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	3500 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	24 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	13 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	30
Detail Type	Platform
Detail ID	PLAT-1
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	43200 cm2
Platform Weight	4414.8 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	1500 mm.
Platform Height	1200 mm.
Platform Clearance or End Offset	230 mm.
Platform Force Coefficient	1.2
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	0 mm.
Unit Weight of Ladder	0 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.11981E-05 kg./cm3

Element From Node	30
Detail Type	Nozzle
Detail ID	MH124"
Dist. from "FROM" Node / Offset dist	1000 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	40
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	502.14 Kg
Grade of Attached Flange	GR 1.1



Nozzle Matl	SA-516 70
Element From Node	30
Detail Type	Nozzle
Detail ID	N14"
Dist. from "FROM" Node / Offset dist	350 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	23.28 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell 2
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	3500 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	24 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	13 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Platform
Detail ID	PLAT-2
Dist. from "FROM" Node / Offset dist	1000 mm.
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	43200 cm2
Platform Weight	2207.4 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	1500 mm.
Platform Height	1200 mm.
Platform Clearance or End Offset	230 mm.
Platform Force Coefficient	1.2
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	0 mm.
Unit Weight of Ladder	0 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.11981E-05 kg./cm3

Element#	5/7
Element From Node	50
Element To Node	60



Element Type	Cylinder
Description	Shell 2
Distance "FROM" to "TO"	2000 mm.
Inside Diameter	3500 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	24 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	13 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2000 mm.
Liquid Density	0.11981E-05 kg./cm3
Element From Node	50
Detail Type	Weight
Detail ID	Lifting wt
Dist. from "FROM" Node / Offset dist	1261 mm.
Miscellaneous Weight	1000 Kg
Offset from Element Centerline	0 mm.
Element From Node	50
Detail Type	Weight
Detail ID	7% margin
Dist. from "FROM" Node / Offset dist	1261 mm.
Miscellaneous Weight	2280 Kg
Offset from Element Centerline	0 mm.
Element From Node	50
Detail Type	Weight
Detail ID	Cleat
Dist. from "FROM" Node / Offset dist	1261 mm.
Miscellaneous Weight	1260 Kg
Offset from Element Centerline	0 mm.
Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell 3
Distance "FROM" to "TO"	2000 mm.
Inside Diameter	3500 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	24 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	13 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	60



Detail Type	Platform	
Detail ID	PLAT-3	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	43200	cm2
Platform Weight	2207.4	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	1500	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	230	mm.
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	0	mm.
Unit Weight of Ladder	0	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	2000	mm.
Liquid Density	0.11981E-05	kg./cm3

Element From Node	60	
Detail Type	Weight	
Detail ID	Davit	
Dist. from "FROM" Node / Offset dist	1800	mm.
Miscellaneous Weight	800	Kgf
Offset from Element Centerline	1300	mm.

Element#	7/7	
Element From Node	70	
Element To Node	80	
Element Type	Elliptical	
Description	Top D end	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	3500	mm.
Element Thickness	20	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	24	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	13	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	TOP PLT	
Dist. from "FROM" Node / Offset dist	1100	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	147600	cm2
Platform Weight	15919	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	9550	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	4775	mm.



Platform Force Coefficient	1.0	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	11000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	9550	mm.

Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	925	mm.
Liquid Density	0.11981E-05	kg./cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N43"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	21.173	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N24"	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	27.254	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N53"	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.79	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N63"	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	25.987	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	70	
Detail Type	Weight	
Detail ID	PSV-1	
Dist. from "FROM" Node / Offset dist	920	mm.



Miscellaneous Weight	5500	Kgf
Offset from Element Centerline	4775	mm.
Element From Node	70	
Detail Type	Weight	
Detail ID	PSV-2	
Dist. from "FROM" Node / Offset dist	920	mm.
Miscellaneous Weight	5500	Kgf
Offset from Element Centerline	4775	mm.
Element From Node	70	
Detail Type	For./Mom.	
Detail ID	N2FxFz	
Dist. from "FROM" Node / Offset dist	800	mm.
Force in X Direction	2242.9	Kgf
Force in Y Direction	-1513.6	Kgf
Force in Z Direction	2242.9	Kgf
Moment about X Axis	4484.7	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	3588.2	Kg-m.
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Bottom D end		...	2000	...	2000
Bottom D end		...	2050	...	50
Shell 1		...	5050	...	3000
Shell 2		...	8050	...	3000
Shell 2		...	10050	...	2000
Shell 3		...	12050	...	2000
Top D end		...	12100	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom D end

Elliptical Head Design Information:					
Design Pressure	P	13.001 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	164.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm^2
Inside Diameter	Di	3500.000 mm.	Actual Stress		120.9 N./mm^2
Spec. Min. Thk t _e , or s _a		20.0000 mm.	Spec. Nominal Thk.		24.000 mm.
Surface Area		142028.6 cm^2	SG of Contents		0.001
Element Volume		6094472.0 cm3	Weight of Contents		7.3 Kg
Empty Weight		2867.7 Kg	Operating Weight		2875.0 Kg
			Comp. All. Ope. Stress		90.7 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.875 mm.

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

tr = 12.5235 mm.

t = tr + ci + co
 = 12.5235 + 3.0 + 0.0
 = 15.5235 mm. - see below for the derivation

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

h = D/(2 • Aspect Ratio)
 = 3500.0/(2 • 2.0)
 = 875.000 mm.

k = D/(2 • h)
 = 3500.0/(2 • 875.0)
 = 2.000

r = D(0.5/k - 0.08) + ca
 = 3500.0(0.5/2.0 - 0.08) + 3.0
 = 598.000 mm.

L = D(0.44 • k + 0.02) + ca
 = 3500.0(0.44 • 2.0 + 0.02) + 3.0
 = 3153.000 mm.

Step 3: Geometric Constants [βTh, φTh, Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: t = 12.5235 mm., P = 13.0014 bars

Angle [βTh]:

= acos[(0.5 • D - r)/(L - r)]
 = acos[(0.5 • 3500.0 - 598.0)/(3153.0 - 598.0)]
 = 1.1017 radians - Equation 4.3.8



$$\begin{aligned}\phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(3153.0 \cdot 12.524) / 598.0} \\ &= 0.3323 \text{ radians - Equation 4.3.9}\end{aligned}$$

$\phi Th < \beta Th$, Thus:

$$\begin{aligned}Rth &= (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r \\ &= (0.5 \cdot 3506.0 - 598.0) / \cos(1.102 - 0.332) + 598.0 \\ &= 2205.9517 \text{ mm. - Equation 4.3.10}\end{aligned}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$r/D > 0.08$, Thus:

$$\begin{aligned}C1 &= 0.692(r/D) + 0.605 \\ &= 0.692(598.0/3506.0) + 0.605 \\ &= 0.723 \text{ - Equation 4.3.13}\end{aligned}$$

$$\begin{aligned}C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(598.0/3506.0) \\ &= 1.0165, \text{ Equation 4.3.15}\end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [Peth]:

$$\begin{aligned}&= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r)) \\ &= 0.723 \cdot 200199.344 \cdot 12.524^2 / (1.017 \cdot 2205.952(0.5 \cdot 2205.952 - 598.0)) \\ &= 200.48 \text{ bars, Equation 4.3.16}\end{aligned}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 246.0 \text{ N/mm}^2$, Not in the creep range - 4.3.6.1(f)

$$\begin{aligned}Py &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 246.0 \cdot 12.524 / (1.017 \cdot 2205.952(0.5 \cdot 2205.952 / 598.0 - 1)) \\ &= 16.269 \text{ bars, Equation 4.3.17}\end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 200.475/16.269 = 12.323$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$\begin{aligned}&= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py \\ &= ((0.77508 \cdot 12.323 - 0.20354 \cdot 12.323^2 + 0.019274 \cdot 12.323^3) / \\ &\quad (1 + 0.19014 \cdot 12.323 - 0.089534 \cdot 12.323^2 + 0.0093965 \cdot 12.323^3)) \cdot 16.269 \\ &= 32.65 \text{ bars, Equation 4.3.19}\end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 32.65/1.5 = 21.76 \text{ bars - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$\begin{aligned}&= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 164.0 \cdot 1.0 / (3153.0/12.524 + 0.5) \\ &= 13.0 \text{ bars, Equation 4.3.22}\end{aligned}$$

Step 10: Compute final pressure [Pa]:

$$\begin{aligned}&= \min(Pak, Pac) = \min(21.764, 13.001) \\ &= 13.0 \text{ bars, Equation 4.3.23: (same as original design pressure)}\end{aligned}$$

Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned}&= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 164.0 \cdot 1.0 / (3153.0/17.0 + 0.5) \\ &= 17.636 \text{ bars}\end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 17.6362 - 0.001 = 17.6348 \text{ bars}$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 22.1502 \text{ bars}$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 120.9 \text{ N/mm}^2$$

Required Thickness of Straight Flange = 16.953 mm.

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.965 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$ 19.474 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:
 $= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$
 $= \text{square root}(4 \cdot 142028.6 / 3.14159)$
 $= 4252.486 \text{ mm.}$

MDMT Calculations in the Knuckle Portion:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 4 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 9 °C
 Temperature reduction per Graph 3.12 : 5 °C
 The computed stress ratio is : 0.882

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (14.99 \cdot 1.0) / (20.0 - 3.0 - 0.0)$
 $= 0.882$

MDMT Calculations in the Head Straight Flange:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 4 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 13 °C
 Temperature reduction per Graph 3.12 : 8 °C
 The computed stress ratio is : 0.796

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (16.708 \cdot 1.0) / (24.0 - 3.0 - 0.0)$
 $= 0.796$

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	16.599 bars
Governing Thk.	24.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.796	Basic MDMT	13 °C
		Temperature Difference	8 °C
		Computed Min. Temp.	4 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.875 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	13.001 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	164.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm^2
Inside Diameter	Di	3500.000 mm.	Actual Stress		127.7 N./mm^2
Spec. Min. Thk t _e , or s _a		24.0000 mm.	Spec. Nominal Thk.		24.000 mm.
Surface Area		334391.2 cm^2	SG of Contents		0.001
Element Volume		28868552.0 cm3	Weight of Contents		34.6 Kg
Empty Weight		6178.0 Kg	Operating Weight		6212.7 Kg
			Comp. All. Ope. Stress		96.4 N./mm^2

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3506.0 \cdot (\exp(13.001/(164.0 \cdot 0.85)) - 1) + 3.0 + 0.0$
 $= 16.427 + 3.0 + 0.0 = 19.427 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 164.0 \cdot 0.85 \cdot \ln((2 \cdot 21.0 + 3506.0) / 3506.0) - 0.001$
 $= 16.599 - 0.001 = 16.5979 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 174.997 \cdot 0.85 \cdot \ln((2 \cdot 24.0 + 3500.0) / 3500.0)$
 $= 20.2599 \text{ bars}$

Computed Stress Intensity at Design Pressure [S_i]:
 $= P \cdot D / (E (D_o - D))$
 $= 13.001 \cdot 3506.0 / (0.85 (3548.0 - 3506.0))$
 $= 127.6890 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [S_m]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 13.001 / (0.85 \cdot \ln((2 \cdot 21.0 + 3506.0) / 3506.0))$
 $= 128.4528 \text{ N./mm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 0.681 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 4 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 13 °C
 Temperature reduction per Graph 3.12 : 8 °C



The computed stress ratio is : 0.796

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (19.673 • 0.85)/(24.0 - 3.0 - 0.0)
= 0.796

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	16.599 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm.	Basic MDMT	13 °C
Stress/Thk. Ratio	0.796	Temperature Difference	8 °C
		Computed Min. Temp.	4 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	13.001 bars	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	164.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm^2
Inside Diameter	Di	3500.000 mm.	Actual Stress	127.7 N./mm^2	
Spec. Min. Thk t,e, or sa		24.0000 mm.	Spec. Nominal Thk.	24.000 mm.	
Surface Area		334391.2 cm^2	SG of Contents	0.001	
Element Volume		28868552.0 cm3	Weight of Contents	34.6 Kg	
Empty Weight		6178.0 Kg	Operating Weight	6212.7 Kg	
			Comp. All. Ope. Stress	96.4 N./mm^2	

Radiography Information:			
...			...
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 3506.0 \cdot (\exp(13.001/(164.0 \cdot 0.85)) - 1) + 3.0 + 0.0$$

$$= 16.4266 + 3.0 + 0.0 = 19.4266 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 164.0 \cdot 0.85 \cdot \ln((2 \cdot 21.0 + 3506.0) / 3506.0) - 0.001$$

$$= 16.599 - 0.001 = 16.5982 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 174.997 \cdot 0.85 \cdot \ln((2 \cdot 24.0 + 3500.0) / 3500.0)$$

$$= 20.2599 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 13.001 \cdot 3506.0 / (0.85 (3548.0 - 3506.0))$$

$$= 127.6856 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 13.001 / (0.85 \cdot \ln((2 \cdot 21.0 + 3506.0) / 3506.0))$$

$$= 128.4494 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 0.681 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-16 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-5 °C
Temperature reduction per Graph 3.13	:	11 °C



The computed stress ratio is : 0.796

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (19.673 • 0.85)/(24.0 - 3.0 - 0.0)
= 0.796

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	16.599 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.796	Temperature Difference	11 °C
		Computed Min. Temp.	-16 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	13.001 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	164.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm^2
Inside Diameter	Di	3500.000 mm.	Actual Stress		127.7 N./mm^2
Spec. Min. Thk t,e, or sa		24.0000 mm.	Spec. Nominal Thk.		24.000 mm.
Surface Area		222927.4 cm^2	SG of Contents		0.001
Element Volume		19245702.0 cm3	Weight of Contents		23.1 Kgf
Empty Weight		4118.7 Kgf	Operating Weight		4141.8 Kgf
			Comp. All. Ope. Stress		96.4 N./mm^2

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3506.0 \cdot (\exp(13.001/(164.0 \cdot 0.85)) - 1) + 3.0 + 0.0$
 $= 16.4262 + 3.0 + 0.0 = 19.4262 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 164.0 \cdot 0.85 \cdot \ln((2 \cdot 21.0 + 3506.0)/3506.0) - 0.001$
 $= 16.599 - 0.001 = 16.5986 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 174.997 \cdot 0.85 \cdot \ln((2 \cdot 24.0 + 3500.0)/3500.0)$
 $= 20.2599 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 13.001 \cdot 3506.0 / (0.85(3548.0 - 3506.0))$
 $= 127.6821 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D)/D))$
 $= 13.001 / (0.85 \cdot \ln((2 \cdot 21.0 + 3506.0)/3506.0))$
 $= 128.4459 \text{ N./mm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.681 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 4 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 13 °C
 Temperature reduction per Graph 3.12 : 8 °C



The computed stress ratio is : 0.796

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (19.672 • 0.85)/(24.0 - 3.0 - 0.0)
= 0.796

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	16.598 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm.	Basic MDMT	13 °C
Stress/Thk. Ratio	0.796	Temperature Difference	8 °C
		Computed Min. Temp.	4 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 3

Cylindrical Shell Design Information:					
Design Pressure	P	13.000 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	164.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm^2
Inside Diameter	Di	3500.000 mm.	Actual Stress		127.7 N./mm^2
Spec. Min. Thk t,e, or sa		24.0000 mm.	Spec. Nominal Thk.		24.000 mm.
Surface Area		222927.4 cm^2	SG of Contents		0.001
Element Volume		19245702.0 cm3	Weight of Contents		23.1 Kg
Empty Weight		4118.7 Kg	Operating Weight		4141.8 Kg
			Comp. All. Ope. Stress		96.4 N./mm^2

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 3506.0 \cdot (\exp(13.0/(164.0 \cdot 0.85)) - 1) + 3.0 + 0.0$$

$$= 16.4258 + 3.0 + 0.0 = 19.4258 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 164.0 \cdot 0.85 \cdot \ln((2 \cdot 21.0 + 3506.0) / 3506.0) - 0.$$

$$= 16.599 - 0. = 16.5988 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 174.997 \cdot 0.85 \cdot \ln((2 \cdot 24.0 + 3500.0) / 3500.0)$$

$$= 20.2599 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 13.0 \cdot 3506.0 / (0.85 (3548.0 - 3506.0))$$

$$= 127.6798 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 13.0 / (0.85 \cdot \ln((2 \cdot 21.0 + 3506.0) / 3506.0))$$

$$= 128.4435 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 0.681 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -16 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : -5 °C
 Temperature reduction per Graph 3.13 : 11 °C



The computed stress ratio is : 0.796

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (19.672 • 0.85)/(24.0 - 3.0 - 0.0)
= 0.796

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	16.598 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	24.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.796	Temperature Difference	11 °C
		Computed Min. Temp.	-16 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Top D end

Elliptical Head Design Information:					
Design Pressure	P	13.000 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	164.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm^2
Inside Diameter	Di	3500.000 mm.	Actual Stress		120.9 N./mm^2
Spec. Min. Thk t _e , or s _a		20.0000 mm.	Spec. Nominal Thk.		24.000 mm.
Surface Area		142028.6 cm^2	SG of Contents		0.001
Element Volume		6094472.0 cm3	Weight of Contents		7.3 Kg
Empty Weight		2867.7 Kg	Operating Weight		2875.0 Kg
			Comp. All. Ope. Stress		90.7 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.875 mm.

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

tr = 12.5223 mm.

t = tr + ci + co
 = 12.5223 + 3.0 + 0.0
 = 15.5223 mm. - see below for the derivation

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

h = D/(2 • Aspect Ratio)
 = 3500.0/(2 • 2.0)
 = 875.000 mm.

k = D/(2 • h)
 = 3500.0/(2 • 875.0)
 = 2.000

r = D(0.5/k - 0.08) + ca
 = 3500.0(0.5/2.0 - 0.08) + 3.0
 = 598.000 mm.

L = D(0.44 • k + 0.02) + ca
 = 3500.0(0.44 • 2.0 + 0.02) + 3.0
 = 3153.000 mm.

Step 3: Geometric Constants [βTh, φTh, Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: t = 12.5223 mm., P = 13.0 bars

Angle [βTh]:

= acos[(0.5 • D - r)/(L - r)]
 = acos[(0.5 • 3500.0 - 598.0)/(3153.0 - 598.0)]
 = 1.1017 radians - Equation 4.3.8

$$\begin{aligned}\phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(3153.0 \cdot 12.522) / 598.0} \\ &= 0.3323 \text{ radians - Equation 4.3.9}\end{aligned}$$

$\phi Th < \beta Th$, Thus:

$$\begin{aligned}Rth &= (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r \\ &= (0.5 \cdot 3506.0 - 598.0) / \cos(1.102 - 0.332) + 598.0 \\ &= 2205.9775 \text{ mm. - Equation 4.3.10}\end{aligned}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$r/D > 0.08$, Thus:

$$\begin{aligned}C1 &= 0.692(r/D) + 0.605 \\ &= 0.692(598.0/3506.0) + 0.605 \\ &= 0.723 \text{ - Equation 4.3.13}\end{aligned}$$

$$\begin{aligned}C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(598.0/3506.0) \\ &= 1.0165, \text{ Equation 4.3.15}\end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [Peth]:

$$\begin{aligned}&= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r)) \\ &= 0.723 \cdot 200199.344 \cdot 12.522^2 / (1.017 \cdot 2205.978(0.5 \cdot 2205.978 - 598.0)) \\ &= 200.43 \text{ bars, Equation 4.3.16}\end{aligned}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 246.0 \text{ N/mm}^2$, Not in the creep range - 4.3.6.1(f)

$$\begin{aligned}Py &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 246.0 \cdot 12.522 / (1.017 \cdot 2205.978(0.5 \cdot 2205.978 / 598.0 - 1)) \\ &= 16.266 \text{ bars, Equation 4.3.17}\end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 200.428/16.266 = 12.322$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$\begin{aligned}&= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py \\ &= ((0.77508 \cdot 12.322 - 0.20354 \cdot 12.322^2 + 0.019274 \cdot 12.322^3) / \\ &\quad (1 + 0.19014 \cdot 12.322 - 0.089534 \cdot 12.322^2 + 0.0093965 \cdot 12.322^3)) \cdot 16.266 \\ &= 32.64 \text{ bars, Equation 4.3.19}\end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 32.64/1.5 = 21.76 \text{ bars - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$\begin{aligned}&= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 164.0 \cdot 1.0 / (3153.0/12.522 + 0.5) \\ &= 13.0 \text{ bars, Equation 4.3.22}\end{aligned}$$

Step 10: Compute final pressure [Pa]:

$$\begin{aligned}&= \min(Pak, Pac) = \min(21.761, 13.0) \\ &= 13.0 \text{ bars, Equation 4.3.23: (same as original design pressure)}\end{aligned}$$

Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned}&= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 164.0 \cdot 1.0 / (3153.0/17.0 + 0.5) \\ &= 17.636 \text{ bars}\end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 17.6362 - 0. = 17.6361 \text{ bars}$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 22.1502 \text{ bars}$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 120.9 \text{ N/mm}^2$$

Required Thickness of Straight Flange = 16.952 mm.

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.965 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$ 19.474 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:
 $= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$
 $= \text{square root}(4 \cdot 142028.6 / 3.14159)$
 $= 4252.486 \text{ mm.}$

MDMT Calculations in the Knuckle Portion:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 4 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 9 °C
 Temperature reduction per Graph 3.12 : 5 °C
 The computed stress ratio is : 0.882

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (14.989 \cdot 1.0) / (20.0 - 3.0 - 0.0)$
 $= 0.882$

MDMT Calculations in the Head Straight Flange:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 4 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 13 °C
 Temperature reduction per Graph 3.12 : 8 °C
 The computed stress ratio is : 0.796

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (16.707 \cdot 1.0) / (24.0 - 3.0 - 0.0)$
 $= 0.796$

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	16.598 bars
Governing Thk.	24.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.796	Basic MDMT	13 °C
		Temperature Difference	8 °C
		Computed Min. Temp.	4 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.875 mm.

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Hydrostatic Test Pressure Results:

User Defined Hydrostatic Test Pressure at High Point 1.000 bars

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	10.6	247.0	0.043	1.34
Shell 1	11.6	247.0	0.047	1.34
Shell 2	11.6	247.0	0.047	1.34
Shell 2	11.6	247.0	0.047	1.34
Shell 3	11.6	247.0	0.047	1.34
Top D end	10.6	247.0	0.043	1.34

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_3"	Nozzle	161.00	151.00	1.066
N3_3"	Pad	175.00	164.00	1.067
MH1_24"	Nozzle	175.00	164.00	1.067
MH1_24"	Pad	175.00	164.00	1.067
N1_4"	Nozzle	161.00	151.00	1.066
N1_4"	Pad	175.00	164.00	1.067
N4_3"	Nozzle	161.00	151.00	1.066
N4_3"	Pad	175.00	164.00	1.067
N2_4"	Nozzle	161.00	151.00	1.066
N2_4"	Pad	175.00	164.00	1.067
N5_3"	Nozzle	161.00	151.00	1.066
N5_3"	Pad	175.00	164.00	1.067
N6_3"	Nozzle	161.00	151.00	1.066
N6_3"	Pad	175.00	164.00	1.067

Minimum 1.066

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	175.00	164.00	1.067
Shell 1	175.00	164.00	1.067
Shell 2	175.00	164.00	1.067
Shell 2	175.00	164.00	1.067
Shell 3	175.00	164.00	1.067
Top D end	175.00	164.00	1.067

Minimum 1.067

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3_3"	0.54	228.00	0.002
MH1_24"	2.61	247.00	0.011
N1_4"	0.62	228.00	0.003
N4_3"	0.54	228.00	0.002
N2_4"	0.72	228.00	0.003
N5_3"	0.54	228.00	0.002
N6_3"	0.54	228.00	0.002



Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm^2
10	20	2000	...	...	No Calc	No Calc
20	30	No Calc	3540	17	No Calc	No Calc
30	40	10683.3	3548	21	No Calc	No Calc
40	50	10683.3	3548	21	No Calc	No Calc
50	60	10683.3	3548	21	No Calc	No Calc
60	70	10683.3	3548	21	No Calc	No Calc
70	80	No Calc	3540	17	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	20	No Calc	...	4.42441
30	40	24	No Calc	...	1.83395
40	50	24	No Calc	...	1.83395
50	60	24	No Calc	...	1.83395
60	70	24	No Calc	...	1.83395
70	80	20	No Calc	...	4.42441

Minimum

2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	2000	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	10683.3	No Calc	No Calc	No Calc
40	50	10683.3	No Calc	No Calc	No Calc
50	60	10683.3	No Calc	No Calc	No Calc
60	70	10683.3	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	2054.58	...	1926.52	...	...
20	30	2867.7	6094472	2509.24	6125041	...
30	40	6178.03	28868552	5410.37	28967618	...
40	50	6178.03	28868552	5410.37	28967618	...
50	60	4118.68	19245702	3606.92	19311744	...
60	70	4118.68	19245702	3606.92	19311744	...
70	80	2867.7	6094472	2509.24	6125041	...
Total		28383	0.108E+09	24979	0.109E+09	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Wght	330	...	1000	...	skirt manway wt
20	Liqd	7.3089	...	-437.5	...	Liquid: 20
20	Nozl	51.36	...	-1015	...	N3_3"
30	Plat	4414.76	0.24782E-04	...	...	PLAT-1
30	Liqd	34.6211	...	1500	...	Liquid: 30
30	Nozl	502.141	1875.01	1000	...	MH1_24"
30	Nozl	23.2803	-1850	350	...	N1_4"
40	Plat	2207.38	-1781.7	999.999	...	PLAT-2
40	Liqd	34.6211	...	1500	...	Liquid: 40
50	Liqd	23.0807	...	1000	...	Liquid: 50
50	Wght	1000	...	1260.97	...	Lifting wt
50	Wght	2280	...	1260.97	...	7% margin
50	Wght	1260	...	1260.97	...	Cleat
60	Plat	2207.38	1781.7	...	...	PLAT-3
60	Liqd	23.0807	...	1000	...	Liquid: 60
60	Wght	800	-1300	1800	...	Davit
70	Plat	15919.4	...	1100	...	TOP PLT
70	Liqd	7.3089	...	487.5	...	Liquid: 70
70	Nozl	21.1727	...	1115	...	N4_3"
70	Nozl	27.2545	1000	958.07	...	N2_4"
70	Nozl	18.7898	...	958.07	-1000	N5_3"
70	Nozl	25.9866	-1000	958.07	...	N6_3"
70	Wght	5500	4775	920	...	PSV-1
70	Wght	5500	4702.46	920	-829.17	PSV-2
70	Forc	...	...	800	...	N2_FxFz

Total Weight of Each Detail Type:

Platforms	24748.9
Liquid	130.0
Nozzles	670.0
Weights	16670.0

Sum of the Detail Weights 42218.9 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	28383.4	28383.4	28383.4	28383.4	28383.4	28383.4
Nozzles	670.0	670.0	670.0	670.0	670.0	670.0
Wld Weights	5670.0	5670.0	5670.0	5670.0	5670.0	5670.0
Platforms	...	...	...	24748.9	24748.9	24748.9
Empty Weights	...	...	...	11000.0	11000.0	...
Test Weights	...	11000.0	...	...	...	...
Ope Weights	...	...	...	...	...	11000.0

Ope. Liquid	...	...	...	...	...	130.0
Test Liquid	...	108351.3	...	...	...	...
Totals	34723.4	154074.7	34723.4	70472.3	70472.3	70602.3

Field Installation Options:

Platforms installed before lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	34723.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	154074.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	34723.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	70472.3 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	70472.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	70602.3 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	44306.3 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	86008.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	45630.7 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	221796	220791
20	30	142029	138286
30	40	334391	329867
40	50	334391	329867
50	60	222927	219912
60	70	222927	219912
70	80	142029	138286

Total	1620490.8	1596920.6 cm2
Total	1744.3	1718.9 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	2384.58	2384.58	2384.58	...	2384.58
20	30	2919.06	2926.37	9009.82	...	2919.06
30	40	6703.46	11152.8	39969.1	898.467	11118.2
40	50	6178.03	8420.04	37236.3	3932.96	8385.41
50	60	8658.68	8681.76	27892.6	...	8658.68
60	70	4918.68	7149.14	26360	2892.94	7126.06
70	80	13960.9	29887.6	35971.1	52128.3	29880.3

Empty Support Force + the Sum of the Y forces	47237.00	Kgf
Operating Support Force + the Sum of the Y forces	72115.95	Kgf
Hydro Support Force + the Sum of the Y forces	180337.20	Kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	70472.3	70602.3	178824
20	30	68087.7	68217.8	176439
30	40	65168.7	65291.4	167429
40	50	54050.5	54138.6	127460
50	60	45665	45718.5	90223.7



60	70	37006.4	37036.8	62331.1
70	80	29880.3	29887.6	35971.1

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	59852.7	59852.7	59852.7
20	30	59852.7	59852.7	59852.7
30	40	59852.7	59852.7	59852.7
40	50	58954.2	58954.2	58954.2
50	60	55021.2	55021.2	55021.2
60	70	55021.2	55021.2	55021.2
70	80	52128.3	52128.3	52128.3

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3_3"	18.68	19.60	65	150	GR 1.1	...	...	...	...
MH1_24"	18.68	19.60	65	150	GR 1.1	...	...	...	...
N1_4"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N4_3"	18.68	19.60	65	150	GR 1.1	...	...	...	...
N2_4"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N5_3"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N6_3"	48.87	51.00	65	300	GR 1.1	...	...	...	...
Min Rating	18.680	19.600, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 5.79892 Hz.

The Natural Frequency for the Vessel (Ope...) is 5.79565 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	3171.94	8281.09

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	3171.94	8281.09

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
70	800.00	2243.	-1514.	2243.	4485.		3588.



Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 3.667

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 5.796 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (3.667) is ≤ 5 .

Note: Dynamic wind analysis is not required.

Design Wind Speed [Vz]:

$$= \text{Basic Wind Speed} \cdot k_1 \cdot k_2 \cdot k_3 \cdot k_4 \quad (\text{for the 1st element})$$

$$= 50.0 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 1.15$$

$$= 57.496 \text{ meters/s}$$

Design Wind Pressure per 4.5.2 [pd]:

$$= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$$

$$= \max(0.7 \cdot 1983.503, 1.0 \cdot 0.84 \cdot 1.0 \cdot 0.6 \cdot 57.5^2)$$

$$= 1674.5 \text{ N/m}^2$$

Element Wind Load (operating condition):

$$= \text{Wind Area} \cdot C_f \cdot p_d$$

$$= 8.5 \cdot 0.756 \cdot 1674.517$$

$$= 10697.6 \text{ N}$$

From	Height mm.	k1	k2	k3	Vz m/sec	Cf
10	1300.67	1.0000	1.0000	1.0000	57.50	0.7556
20	2325.00	1.0000	1.0000	1.0000	57.50	0.7556
30	3850.00	1.0000	1.0000	1.0000	57.50	0.7556
40	6850.00	1.0000	1.0000	1.0000	57.50	0.7556
50	9350.00	1.0000	1.0000	1.0000	57.50	0.7556
60	11350.00	1.0000	1.0135	1.0000	58.27	0.7556
70	12752.96	1.0000	1.0275	1.0000	59.08	0.7556

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency



D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: 0.0/12100.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[11.63/39.7^2] \cdot 10^4 = 73.776$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.400E-06$ no chance. [Vp]:

= $W / (L \cdot Dr^2)$

= $67198 / (12100.0 \cdot 3547.934^2)$

= 0.44119E-06

Since V_p is > 0.400E-06 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf	Cf
PLAT-1	43200.00	2350.00	170.83	737.97	1.20
PLAT-2	43200.00	6350.00	170.83	737.97	1.20
PLAT-3	43200.00	10350.00	173.15	748.00	1.20
TOP PLT	147600.00	13450.00	182.37	2691.85	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
skirt manway	0.00	1300.00	170.83	0.00
Lifting wt	0.00	9610.97	171.43	0.00
7% margin	0.00	9610.97	171.43	0.00
Cleat	0.00	9610.97	171.43	0.00
Davit	0.00	12150.00	178.26	0.00
PSV-1	0.00	13270.00	182.18	0.00
PSV-2	0.00	13270.00	182.18	0.00

The Natural Frequency for the Vessel (Ope...) is 5.79565 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	1300.67	4227.6	84552	170.826	1200.01
20	30	2325	4248	2124	170.826	30.1451
30	40	3850	4257.6	127728	170.826	2308.29
40	50	6850	4257.6	127728	170.826	2565.31
50	60	9350	4257.6	85152	170.826	1208.53



60	70	11350	4257.6	85152	175.469	1991.68
70	80	12753	4248	31984.5	180.361	3602.16

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Input Values:

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Computed Natural Frequencies (OPE): 5

Mode	Freq (Hz)	Freq (Rad/Sec)	Period (Sec)
1	5.7957	36.4152	0.1725
2	37.5564	235.9736	0.0266
3	40.7453	256.0100	0.0245
4	89.0733	559.6642	0.0112
5	138.3003	868.9663	0.0072

Mass Participation Factors: 5

Mode	X	Y
1	0.91693913	0.00000000
2	0.00000000	0.00805431
3	0.00408680	0.00000000
4	-0.00023285	-0.00000000
5	0.00002962	0.00000000

Computed EigenVectors, Mass Normalized: 5

Node	Dx	Dy	Rz
Mode # 1	Freq (HZ) = 5.7957		
10	0.000000	0.000000	-0.000000
20	0.006793	0.000000	-0.000093
30	0.007005	0.000000	-0.000094
40	0.021427	-0.000000	-0.000125
50	0.038746	-0.000000	-0.000144
60	0.051151	-0.000000	-0.000151
70	0.063764	0.000000	-0.000153
80	0.064073	0.000000	-0.000153



Mode # 2	Freq (HZ) =	37.5564		
10	0.000000	0.000001	0.000000	
20	0.000000	0.026251	0.000000	
30	0.000000	0.026497	0.000000	
40	-0.000000	0.037762	-0.000000	
50	0.000000	0.047701	0.000000	
60	-0.000000	0.053353	0.000000	
70	0.000000	0.057994	-0.000000	
80	0.000000	0.058058	-0.000000	

Mode # 3	Freq (HZ) =	40.7453		
10	0.000001	-0.000000	-0.000000	
20	0.070712	-0.000000	-0.000057	
30	0.071376	-0.000000	-0.000056	
40	0.086525	0.000000	0.000093	
50	0.060111	0.000000	0.000306	
60	0.022336	0.000000	0.000410	
70	-0.023724	0.000000	0.000448	
80	-0.024773	0.000000	0.000448	

Mode # 4	Freq (HZ) =	89.0733		
10	-0.000001	-0.000000	0.000000	
20	-0.095894	-0.000000	-0.000322	
30	-0.095772	-0.000000	-0.000328	
40	-0.002339	-0.000000	-0.000418	
50	0.078407	-0.000000	-0.000069	
60	0.055086	-0.000000	0.000267	
70	-0.018781	-0.000000	0.000413	
80	-0.020245	-0.000000	0.000413	

Mode # 5	Freq (HZ) =	138.3003		
10	0.000001	0.000000	0.000000	
20	0.066526	0.000000	0.000390	
30	0.065705	0.000000	0.000396	
40	-0.094854	0.000000	0.000128	
50	0.032317	0.000000	-0.000157	
60	0.065612	0.000000	0.000111	
70	-0.009494	0.000000	0.000295	
80	-0.010920	0.000000	0.000295	

Mass Percentages:				
Included(X)	Included(Y)	Added(X)	Added(Y)	
98.78	93.77	1.22	6.23	

NOTE: In the following reports, "Max Contrib" displays the contribution of the Mode/Load combination having the maximum impact on the total (and names that Mode/Load combination).

Restraint Loads:				
Node	Fx(Kgf)	Fy(Kgf)	Mz(Kg-m.)	
10	9965.2	3941.4	106270.9	
Max	9917.6	3936.1	106263.8	
Contrib	1 (X)	2 (Y)	1 (X)	

Element Forces and Moments:				
Node	Fx(Kgf)	Fy(Kgf)	Mz(Kg-m.)	
10	9965.1	3939.6	106270.9	
Max	9917.6	3936.1	106263.8	
Contrib	1 (X)	2 (Y)	1 (X)	
20	9965.1	3939.6	86431.1	
Max	9917.6	3936.1	86428.1	
Contrib	1 (X)	2 (Y)	1 (X)	
20	9905.1	3867.2	86431.1	
Max	9868.2	3865.6	86428.1	



Contrib	1 (X)	2 (Y)	1 (X)
30	9905.1	3867.2	85938.1
Max	9868.2	3865.6	85934.7
Contrib	1 (X)	2 (Y)	1 (X)
30	9726.4	3649.3	85938.1
Max	9712.9	3649.0	85934.7
Contrib	1 (X)	2 (Y)	1 (X)
40	9726.4	3649.3	56840.6
Max	9712.9	3649.0	56795.5
Contrib	1 (X)	2 (Y)	1 (X)
40	9053.3	3224.4	56840.6
Max	9052.5	3219.7	56795.5
Contrib	1 (X)	2 (Y)	1 (X)
50	9053.3	3224.4	29707.8
Max	9052.5	3219.7	29637.6
Contrib	1 (X)	2 (Y)	1 (X)
50	8021.2	2753.5	29707.8
Max	8009.1	2745.9	29637.6
Contrib	1 (X)	2 (Y)	1 (X)
60	8021.2	2753.5	13668.7
Max	8009.1	2745.9	13619.1
Contrib	1 (X)	2 (Y)	1 (X)
60	6758.6	2263.5	13668.7
Max	6734.0	2255.3	13619.1
Contrib	1 (X)	2 (Y)	1 (X)
70	6758.6	2263.5	151.3
Max	6734.0	2255.3	150.8
Contrib	1 (X)	2 (Y)	1 (X)
70	3026.9	1011.5	151.3
Max	3015.4	1007.8	150.8
Contrib	1 (X)	2 (Y)	1 (X)
80	3026.9	1011.5	0.0
Max	3015.4	1007.8	0.0
Contrib	1 (X)	2 (Y)	

Displacements:

Node	Dx(mm.)	Dy(mm.)	Rz(Deg)
10	0.0000	0.0000	0.0000
Max	0.0000	0.0000	0.0000
Contrib	1 (X)	2 (Y)	1 (X)
20	0.1584	0.0054	0.0049
Max	0.1582	0.0054	0.0049
Contrib	1 (X)	2 (Y)	1 (X)
30	0.1633	0.0054	0.0049
Max	0.1631	0.0054	0.0049
Contrib	1 (X)	2 (Y)	1 (X)
40	0.4991	0.0077	0.0066
Max	0.4991	0.0077	0.0066
Contrib	1 (X)	2 (Y)	1 (X)
50	0.9024	0.0098	0.0076
Max	0.9024	0.0098	0.0076
Contrib	1 (X)	2 (Y)	1 (X)
60	1.1913	0.0109	0.0079
Max	1.1913	0.0109	0.0079
Contrib	1 (X)	2 (Y)	1 (X)
70	1.4851	0.0119	0.0080
Max	1.4851	0.0119	0.0080
Contrib	1 (X)	2 (Y)	1 (X)



80	1.4923	0.0119	0.0080
Max	1.4923	0.0119	0.0080
Contrib	1 (X)	2 (Y)	1 (X)



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm.	Cumulative Shr Wind Cas Kg-f	Cumulative Shr Eqk Cas Kg-f	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	3171.94	9965.06	46473	46473
20	30	2025	3171.94	9905.11	40129.6	40129.6
30	40	3550	3171.94	9726.36	39971.1	39971.1
40	50	6550	3171.94	9053.29	30456.6	30456.6
50	60	9050	3171.94	8021.22	20943.6	20943.6
60	70	11050	3171.94	6758.55	14603.8	14603.8
70	80	12075	3171.94	3026.87	8270.33	8270.33

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kg-f	Earthquake Shear Kg-f	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	12906.1	9965.06	104063	106271
20	30	2025	11706.1	9905.11	79450.5	86431.1
30	40	3550	11676	9726.36	78865.9	85938.1
40	50	6550	9367.68	9053.29	47299.8	56840.6
50	60	9050	6802.37	8021.22	23044.2	29707.8
60	70	11050	5593.85	6758.55	10647.8	13668.7
70	80	12075	3602.16	3026.87	1451.57	151.347

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	3171.94	0.0089206	0.034852	0.34021E-04
20	30	3171.94	0.035704	0.036561	0.34355E-04
30	40	3171.94	0.093894	0.16187	0.48546E-04
40	50	3171.94	0.23905	0.324	0.58903E-04
50	60	3171.94	0.38424	0.44687	0.63679E-04
60	70	3171.94	0.51146	0.57759	0.66753E-04
70	80	3171.94	0.57925	0.58092	0.66753E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 60.500 Actual deflection : 0.581 mm.

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	12906.1	0.019757	0.075743	0.72494E-04
20	30	11706.1	0.077559	0.079384	0.73158E-04
30	40	11676	0.20027	0.34014	0.98544E-04
40	50	9367.68	0.49447	0.65927	0.00011257
50	60	6802.37	0.77325	0.88948	0.00011707
60	70	5593.85	1.00715	1.12559	0.00011862
70	80	3602.16	1.12855	1.13152	0.00011862

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	439.843	2749.02
20	30	441.965	2762.28
30	40	442.964	2768.52
40	50	442.964	2768.52
50	60	442.964	2768.52
60	70	442.964	2768.52
70	80	441.965	2762.28

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 60.500 Actual deflection : 1.132 mm.

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 60.500 Actual deflection : 1.712 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	881.671	716.674	77147160	62763236
20	30	2211.68	1881.53	193534352	164923472
30	40	2657.05	2326.9	232513328	203966976
40	50	2657.05	2326.9	232513328	203966976
50	60	2657.05	2326.9	232513328	203966976
60	70	2657.05	2326.9	232513328	203966976
70	80	2211.68	1881.53	193534352	164923472

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	164	246.996	-46.2069	-56.87
20	30	164	246.996	-90.7214	-95.2081
30	40	164	246.996	-96.435	-99.8127
40	50	164	246.996	-96.435	-99.8127
50	60	164	246.996	-96.435	-99.8127
60	70	164	246.996	-96.435	-99.8127
70	80	164	246.996	-90.7214	-95.2081

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	66.707	...	4.3504
30	40	53.9393	...	3.62119
40	50	53.9393	...	3.62119
50	60	53.9393	...	3.62119
60	70	53.9393	...	3.62119
70	80	66.707	...	4.3504

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-9.64329	-9.66108	...	9.35187	9.35187
20	30	-3.54884	-3.54884	...	3.55894	3.55894
30	40	-2.74657	-2.74657	...	2.87769	2.87769
40	50	-2.27799	-2.27799	...	2.83449	2.83449
50	60	-1.92458	-1.92458	...	2.6454	2.6454
60	70	-1.55965	-1.55965	...	2.6454	2.6454
70	80	-1.55741	-1.55741	...	3.09964	3.09964

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	16.2597	16.6046	...	...
20	30	...	4.72426	5.13935	...	...
30	40	...	3.79184	4.13186	...	...
40	50	...	2.27415	2.73287	...	...
50	60	...	1.10796	1.42834	...	...
60	70	...	0.51194	0.65719	...	...
70	80	...	0.086313	0.008993	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	9.95932
20	30	...	...	...	3.05284
30	40	...	...	...	2.4537
40	50	...	...	...	1.58518
50	60	...	...	...	0.78302
60	70	...	...	...	0.33857
70	80	...	...	...	0.0044161

Longitudinal Stresses due to Applied Axial Forces:



From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	-0.20712	-0.7462
20	30	-0.078891	-0.28046
30	40	-0.063791	-0.21759
40	50	-0.063791	-0.19968
50	60	-0.063791	-0.17984
60	70	-0.063791	-0.15919
70	80	-0.078891	-0.13161

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	7.26131	7.26131	5.90745	5.90745
20	30	2.38618	2.38618	2.03342	2.03342
30	40	1.92179	1.92179	1.68585	1.68585
40	50	1.46434	1.46434	1.28456	1.28456
50	60	1.00696	1.00696	0.88333	0.88333
60	70	0.70214	0.70214	0.61594	0.61594
70	80	0.49177	0.49177	0.41907	0.41907

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads. The stresses are combined per equation 4.3.44 to determine the overall stress intensity. The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	42.72	164.00	-42.72	64.41	0.2605	0.6633	0.3850
20	14.30	164.00	-14.30	87.07	0.0872	0.1642	0.4242
30	11.40	164.00	-11.40	110.43	0.0695	0.1032	0.0742
40	8.91	164.00	-8.91	110.43	0.0544	0.0807	0.0568
50	6.75	164.00	-6.75	110.43	0.0412	0.0611	0.0411
60	5.48	164.00	-5.48	110.43	0.0334	0.0496	0.0333
70	5.31	164.00	-5.31	85.34	0.0324	0.0623	0.0696

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	36.96	164.00	-36.96	64.41	0.2254	0.5739	0.3112
20	12.83	164.00	-12.83	86.27	0.0782	0.1487	0.3613
30	10.22	164.00	-10.22	110.43	0.0623	0.0925	0.0626
40	8.36	164.00	-8.36	110.43	0.0510	0.0757	0.0509
50	6.54	164.00	-6.54	110.43	0.0399	0.0592	0.0607



60	5.40	164.00	-5.40	110.43	0.0330	0.0489	0.0690
70	5.28	164.00	-5.28	85.31	0.0322	0.0619	0.0687

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	42.74	164.00	-42.74	64.41	0.2606	0.6636	0.3850
20	14.30	164.00	-14.30	87.06	0.0872	0.1642	0.4244
30	11.40	164.00	-11.40	110.43	0.0695	0.1032	0.0742
40	8.91	164.00	-8.91	110.43	0.0544	0.0807	0.0568
50	6.75	164.00	-6.75	110.43	0.0412	0.0611	0.0411
60	5.48	164.00	-5.48	110.43	0.0334	0.0496	0.0333
70	5.31	164.00	-5.31	85.34	0.0324	0.0623	0.0696

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	43.63	164.00	-43.63	64.41	0.2660	0.6773	0.3890
20	14.91	164.00	-14.91	87.04	0.0909	0.1713	0.4346
30	11.90	164.00	-11.90	110.43	0.0725	0.1077	0.0771
40	9.51	164.00	-9.51	110.43	0.0580	0.0861	0.0607
50	7.19	164.00	-7.19	110.43	0.0438	0.0651	0.0607
60	5.72	164.00	-5.72	110.43	0.0349	0.0518	0.0690
70	5.29	164.00	-5.29	85.35	0.0323	0.0620	0.0724

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	56.87	0.0000	-0.0000	0.0166
20	0.00	247.00	0.00	95.21	0.0000	-0.0000	...
30	0.00	247.00	0.00	99.81	0.0000	-0.0000	0.0066
40	0.00	247.00	0.00	99.81	0.0000	-0.0000	0.0092
50	0.00	247.00	0.00	99.81	0.0000	-0.0000	0.0104
60	0.00	247.00	0.00	99.81	0.0000	-0.0000	0.0124
70	0.00	247.00	0.00	95.21	0.0000	-0.0000	...

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	56.87	0.0000	-0.0000	0.0121
20	0.00	247.00	0.00	95.21	0.0000	-0.0000	...
30	0.00	247.00	0.00	99.81	0.0000	-0.0000	0.0031
40	0.00	247.00	0.00	99.81	0.0000	-0.0000	0.0031
50	0.00	247.00	0.00	99.81	0.0000	-0.0000	0.0031
60	0.00	247.00	0.00	99.81	0.0000	-0.0000	0.0031
70	0.00	247.00	0.00	95.21	0.0000	-0.0000	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	42.74	164.00	-42.74	64.41	0.2606	0.6636	0.3850
20	105.54	164.00		87.06	0.6436		0.4244
30	112.62	164.00		110.43	0.6867		0.0742
40	112.17	164.00		110.43	0.6840		0.0568
50	111.82	164.00		110.43	0.6818		0.0411
60	111.64	164.00		110.43	0.6807		0.0333
70	105.02	164.00		90.72	0.6404		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	43.63	164.00	-43.63	64.41	0.2660	0.6773	0.3890
20	105.57	164.00		87.04	0.6437		0.4346
30	112.71	164.00		110.43	0.6873		0.0771



40	112.27	164.00	110.43	0.6846	0.0607
50	111.89	164.00	110.43	0.6822	0.0607
60	111.67	164.00	110.43	0.6809	0.0690
70	105.01	164.00	90.72	0.6403	...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	42.74	164.00	-42.74	64.85	0.2606	0.6590	0.3839
20	14.30	164.00	-14.30	87.74	0.0872	0.1629	0.4211
30	11.40	164.00	-11.40	110.55	0.0695	0.1031	0.0741
40	8.91	164.00	-8.91	110.55	0.0544	0.0806	0.0567
50	6.75	164.00	-6.75	110.55	0.0412	0.0610	0.0411
60	5.48	164.00	-5.48	110.55	0.0334	0.0496	0.0333
70	5.31	164.00	-5.31	86.01	0.0324	0.0618	0.0690

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	43.63	164.00	-43.63	64.85	0.2660	0.6727	0.3879
20	14.91	164.00	-14.91	87.72	0.0909	0.1700	0.4313
30	11.90	164.00	-11.90	110.55	0.0725	0.1076	0.0771
40	9.51	164.00	-9.51	110.55	0.0580	0.0860	0.0607
50	7.19	164.00	-7.19	110.55	0.0438	0.0650	0.0603
60	5.72	164.00	-5.72	110.55	0.0349	0.0518	0.0685
70	5.29	164.00	-5.29	86.02	0.0323	0.0615	0.0718

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	56.87	0.0000	-0.0000	0.0166
20	9.24	247.00		95.21	0.0374		...
30	10.28	247.00		99.81	0.0416		0.0066
40	10.28	247.00		99.81	0.0416		0.0092
50	10.28	247.00		99.81	0.0416		0.0104
60	10.28	247.00		99.81	0.0416		0.0124
70	9.24	247.00		95.21	0.0374		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	56.87	0.0000	-0.0000	0.0121
20	9.24	247.00		95.21	0.0374		...
30	10.28	247.00		99.81	0.0416		0.0031
40	10.28	247.00		99.81	0.0416		0.0031
50	10.28	247.00		99.81	0.0416		0.0031
60	10.28	247.00		99.81	0.0416		0.0031
70	9.24	247.00		95.21	0.0374		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	19.00	164.00	-19.00	64.41	0.1158	0.2949	0.1502
20	104.89	164.00		39.86	0.6396		0.0729
30	111.67	164.00		110.43	0.6809		0.0252
40	111.59	164.00		110.43	0.6805		0.0245
50	111.52	164.00		110.43	0.6800		0.0228
60	111.47	164.00		110.43	0.6797		0.0228
70	104.98	164.00		90.72	0.6401		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	7.85	164.00	-7.85	64.41	0.0479	0.1219	0.1223



20	104.75	164.00	39.86	0.6387	...
30	111.27	164.00	110.43	0.6784	0.0221
40	111.22	164.00	110.43	0.6782	0.0183
50	111.18	164.00	110.43	0.6779	0.0155
60	111.15	164.00	110.43	0.6777	0.0125
70	104.81	164.00	90.72	0.6391	...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	19.01	164.00	-19.01	64.41	0.1159	0.2952	0.1505
20	104.89	164.00		39.86	0.6396		0.0737
30	111.67	164.00		110.43	0.6809		0.0252
40	111.59	164.00		110.43	0.6805		0.0245
50	111.52	164.00		110.43	0.6800		0.0228
60	111.47	164.00		110.43	0.6797		0.0228
70	104.98	164.00		90.72	0.6401		...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	19.00	164.00	-19.00	64.41	0.1158	0.2949	0.1502
20	104.89	164.00		39.86	0.6396		0.0729
30	111.67	164.00		110.43	0.6809		0.0252
40	111.59	164.00		110.43	0.6805		0.0245
50	111.52	164.00		110.43	0.6800		0.0228
60	111.47	164.00		110.43	0.6797		0.0228
70	104.98	164.00		90.72	0.6401		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	19.01	164.00	-19.01	64.41	0.1159	0.2952	0.1505
20	7.11	164.00	-7.11	39.86	0.0433	0.1783	0.0737
30	5.62	164.00	-5.62	110.43	0.0343	0.0509	0.0252
40	5.11	164.00	-5.11	110.43	0.0312	0.0463	0.0245
50	4.57	164.00	-4.57	110.43	0.0279	0.0414	0.0228
60	4.21	164.00	-4.21	110.43	0.0256	0.0381	0.0228
70	4.66	164.00	-4.66	85.20	0.0284	0.0547	0.0547

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.02	164.00	-27.02	64.41	0.1648	0.4195	0.1621
20	105.04	164.00		85.06	0.6405		0.2496
30	111.98	164.00		110.43	0.6828		0.0272
40	111.83	164.00		110.43	0.6819		0.0227
50	111.68	164.00		110.43	0.6810		0.0193
60	111.58	164.00		110.43	0.6804		0.0158
70	105.01	164.00		90.72	0.6403		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.02	164.00	-27.02	64.85	0.1648	0.4166	0.1610
20	9.77	164.00	-9.77	85.73	0.0596	0.1140	0.2476
30	7.76	164.00	-7.76	110.55	0.0473	0.0702	0.0271
40	6.78	164.00	-6.78	110.55	0.0413	0.0613	0.0227
50	5.76	164.00	-5.76	110.55	0.0351	0.0521	0.0193
60	5.07	164.00	-5.07	110.55	0.0309	0.0458	0.0157
70	5.28	164.00	-5.28	85.94	0.0322	0.0614	0.0642

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
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10	19.01	164.00	-19.01	64.41	0.1159	0.2952	0.1505
20	104.89	164.00		39.86	0.6396		0.0737
30	111.67	164.00		110.43	0.6809		0.0252
40	111.59	164.00		110.43	0.6805		0.0245
50	111.52	164.00		110.43	0.6800		0.0228
60	111.47	164.00		110.43	0.6797		0.0228
70	104.98	164.00		90.72	0.6401		...

Absolute Maximum of the all of the Stress Ratio's 0.6873

Governing Element: Shell 1
 Governing Load Case 8 : IP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom D end	...	-10.4	9.4	23.9	22.8	+ Tens
Bottom D end	...	-10.4	-9.4	-23.9	-43.6	- Comp
Bottom D end	66.7	-3.8	3.6	7.5	74.0	+ Tens
Bottom D end	66.7	-3.8	-3.6	-7.5	...	- Comp
Shell 1	53.9	-3.0	2.9	6.1	59.9	+ Tens
Shell 1	53.9	-3.0	-2.9	-6.1	...	- Comp
Shell 2	53.9	-2.5	2.8	4.2	58.5	+ Tens
Shell 2	53.9	-2.5	-2.8	-4.2	...	- Comp
Shell 2	53.9	-2.1	2.6	2.4	56.9	+ Tens
Shell 2	53.9	-2.1	-2.6	-2.4	...	- Comp
Shell 3	53.9	-1.7	2.6	1.4	56.2	+ Tens
Shell 3	53.9	-1.7	-2.6	-1.4	...	- Comp
Top D end	66.7	-1.7	3.1	0.5	68.6	+ Tens
Top D end	66.7	-1.7	-3.1	-0.5	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm.
- Fxa - Allowable comp mem stress due to compression with lamc <= 0.15, N./mm²
- Fba - Allowable all comp mem stress due to bending moment, N./mm²
- Fva - Allowable shear stress when only shear is present, N./mm²
- Fha - Allowable hoop comp memb stress under ext pressure alone, N./mm²
- Fxha - Allowable axial comp w/hoop compression and lamc <= 0.15, N./mm²
- Fhxa - Allowable hoop comp w/axial compression and lamc <= 0.15, N./mm²
- Faha - Allowable axial comp w/hoop compression and lamc > 0.15, N./mm²
- Fbha - Allowable axial comp from bending w/hoop compression, N./mm²
- Fhba - Allowable hoop comp w/long compression due to moment, N./mm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						87	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70						85	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						86	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895



50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70						85	6895		6895	6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						87	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70						85	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						87	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70						85	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.175	2000	74	90	58		6895	74	6895	6895	6895
20							6895		6895	6895	6895
30	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
40	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
50	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
60	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.175	2000	74	90	58		6895	74	6895	6895	6895
20							6895		6895	6895	6895
30	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
40	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
50	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
60	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						87	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						87	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.164	2000	64	85	48		6895	64	6895	6895	6895
20						87	6895		6895	6895	6895
30	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.203	10683	110	115	69	15	6895	110	6895	6895	6895



50	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
70						86	6895		6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.164	2000	64	85	48		6895	64	6895	6895	6895
20						87	6895		6895	6895	6895
30	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
70						86	6895		6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.175	2000	74	90	58		6895	74	6895	6895	6895
20							6895		6895	6895	6895
30	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
40	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
50	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
60	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.175	2000	74	90	58		6895	74	6895	6895	6895
20							6895		6895	6895	6895
30	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
40	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
50	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
60	0.207	10683	114	119	74	18	6895	114	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						39	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						39	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						39	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						39	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895



50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						39	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70						85	6895		6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						85	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.164	2000	64	85	48		6895	64	6895	6895	6895
20						85	6895		6895	6895	6895
30	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.203	10683	110	115	69	15	6895	110	6895	6895	6895
70						85	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.165	2000	64	85	48		6895	64	6895	6895	6895
20						39	6895		6895	6895	6895
30	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
40	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
50	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
60	0.205	10683	110	115	69	15	6895	110	6895	6895	6895
70							6895		6895	6895	6895

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Bottom D end

Mean skirt diameter at the critical height is computed later.
 This is a Conical type Support Skirt , Wider at the Top

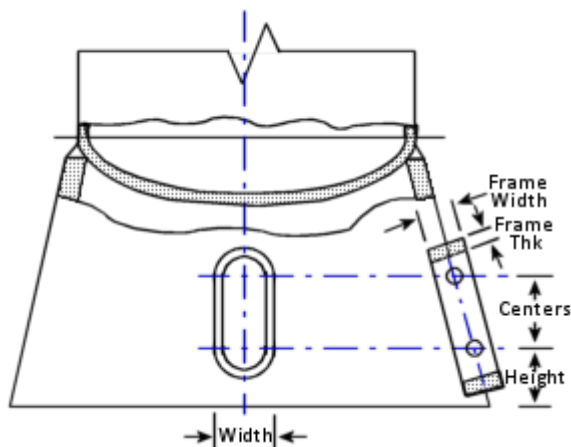
Half Apex Angle of the skirt	a	0.201 deg
Total vessel (empty) weight	Fg	68417.734 Kgf
Head Weight below section 2-2	DeltaFg	2919.063 Kgf
Weight of vessel contents (liquid)	Ff	130.022 Kgf
Vessel Gross Weight	Fg + Ff	68547.758 Kgf
Skirt Mean Diameter Top End	DzTop	3523.500 mm.
Skirt Mean Diameter Bottom End	DzBot	3509.500 mm.
Skirt Analysis Thickness (Corroded)	ez	6.500 mm.
Skirt Height	H	2000.000 mm.
Additional Axial Force at section 1-1	F1	-1513.600 Kgf
Moment at section 1-1	M1	126560.781 Kg-m.
Applied Moment - Bottom of Skirt	Mb	152743.844 Kg-m.
Weight at Bottom of Skirt	W	70602.344 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 3523.000 mm.
Head Corroded Thickness	eb 17.000 mm.

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 878.0 \text{ mm.} \\
 K &= D_i / (2 * h_i) = 3506.0 / (2 * 878.0) = 1.9966 \\
 R &= D_i (0.44 * K + 0.02) = 3506.0 * (0.44 * 1.997 + 0.02) = 3150.1289 \text{ mm.} \\
 r &= D_i ((0.5 / K) - 0.08) = 3506.0 * ((0.5 / 1.997) - 0.08) = 597.52 \text{ mm.}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	11033448.000 cm ⁴
Moment of Inertia	Iyy =	11033448.000 cm ⁴
Section Modulus	W =	62877664.000 mm. ³
Section Metal Area	A =	716.657 cm ²

Analysis of the Section at weakest section Modulus [Z min]:



Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned} &= \sqrt{(\min(lxx, lyy) / \text{Area})} \\ &= \sqrt{(\min(11033448, 11033448) / 716.657)} \\ &= 1240.796 \text{ mm.} \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= rg / L \\ &= 1240.796 / 2000.0 \\ &= 0.620 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 126560.781 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -126560.781 / .25151\text{E}+09 - 70602.344 / 716.65668 \\ &= -14.596 \text{ N./mm}^2 \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +126560.781 / .25151\text{E}+09 - 70602.344 / 716.65668 \\ &= -4.727 \text{ N./mm}^2 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F1 - Fg - Ff + 4 \cdot M1 / Dz \\ &= -1513.6 - 68417.734 - 130.022 + 4 \cdot 126560.781 / 3523.5 \\ &= 73612.0 \text{ Kgf} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F1 - Fg - Ff - 4 \cdot M1 / Dz \\ &= -1513.6 - 68417.734 - 130.022 - 4 \cdot 126560.781 / 3523.5 \\ &= -213734.7 \text{ Kgf} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (73611.992 + 2919.063 + 130.022) / (\pi \cdot 3523.0 \cdot 17.0) + 13.0 \cdot 3523.0 / (4 \cdot 17.0) \\ &= 71.35 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (Fzq + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (-0.2\text{E}+06 + 2919.063 + 130.022) / (\pi \cdot 3523.0 \cdot 17.0) + 13.0 \cdot 3523.0 / (4 \cdot 17.0) \\ &= 56.37 \text{ N./mm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (130.022 + 2919.063) / (\pi \cdot 3523.0 \cdot 17.0) + 13.0 \cdot 3523.0 / (4 \cdot 17.0) \\ &= 67.51 \text{ N./mm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= Fzp / (\pi \cdot Dz \cdot ez) \\ &= 73611.992 / (\pi \cdot 3523.5 \cdot 6.5) \\ &= 10.03 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= Fzq / (\pi \cdot Dz \cdot ez) \\ &= -213734.703 / (\pi \cdot 3523.5 \cdot 6.5) \\ &= -29.13 \text{ N./mm}^2 \end{aligned}$$



16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:
 $= 1 - 0.5 (Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5 (3523.0 + 17.0 - 3509.5 + 6.5) / (597.52 + 17.0)$
 $= 0.9813$

$\gamma = \arccos(\cos(\gamma)) = \arccos(0.981) = 11.102 \text{ deg}$

Dimension [a]:
 $= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$
 $= 0.5 \cdot \sqrt{(17.0^2 + 6.5^2 + 2 \cdot 17.0 \cdot 6.5 \cdot 0.981)}$
 $= 11.706 \text{ mm.}$

Bending moment at verification points p and q [Mp & Mq]:
 $M_p = a \cdot F_{zp} = 11.70 \cdot 73612.0 = 861.71 \text{ Kg-m.}$
 $M_q = a \cdot F_{zq} = 11.70 \cdot -213734.7 = -2502.01 \text{ Kg-m.}$

Correction factor [C]:
 $= 0.63 - 0.057 (eb/ez)^2$
 $= 0.63 - 0.057 (17.0/6.5)^2$
 $= 0.240$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:
 $= C \cdot 6 \cdot M_p / (\pi \cdot Db \cdot eb^2)$
 $= 0.24 \cdot 6 \cdot 861.713 / (\pi \cdot 3523.0 \cdot 17.0^2)$
 $= 3.81 \text{ N./mm}^2$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:
 $= C \cdot 6 \cdot M_q / (\pi \cdot Db \cdot eb^2)$
 $= 0.24 \cdot 6 \cdot -2502.011 / (\pi \cdot 3523.0 \cdot 17.0^2)$
 $= -11.05 \text{ N./mm}^2$

Stress [$\sigma_{3pb(o)}$]:
 $= C \cdot 6 \cdot M_p / (\pi \cdot Dz \cdot ez^2)$
 $= 0.24 \cdot 6 \cdot 861.713 / (\pi \cdot 3523.5 \cdot 6.5^2)$
 $= 26.03 \text{ N./mm}^2$

Stress [$\sigma_{3qb(o)}$]:
 $= C \cdot 6 \cdot M_q / (\pi \cdot Dz \cdot ez^2)$
 $= 0.24 \cdot 6 \cdot -2502.011 / (\pi \cdot 3523.5 \cdot 6.5^2)$
 $= -75.58 \text{ N./mm}^2$

Geometry variables, 16.12-32 to 36:
 $eb/Db = 0.669/138.701 = 0.0048$
 $y = 125 \cdot eb/Db = 125 \cdot 17.0/3523.0 = 0.6032$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [a]:
 $= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$
 $= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.603 + 3.994 = 4.1122$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:
 $= (P + Ph) \cdot Db ((\gamma / \gamma A) \cdot \alpha - 1) / (4 \cdot eb)$
 $= (13.0 + 0.0) \cdot 3523.0 ((11.1/40.0) \cdot 4.112 - 1) / (4 \cdot 17.0)$
 $= 9.52 \text{ N./mm}^2$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:
 $= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$
 $= 71.351 - 3.806 + 9.518$
 $= 77.06 \text{ N./mm}^2$

Stress [σ_{1potot}]:



$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 71.351 + 3.806 - 9.518 \\ &= 65.64 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qitot}] &: \\ &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 56.374 - -11.051 + 9.518 \\ &= 76.94 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qotot}] &: \\ &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 56.374 + -11.051 - 9.518 \\ &= 35.81 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2pitot}] &: \\ &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 67.514 + 3.806 + 9.518 \\ &= 80.84 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2potot}] &: \\ &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 67.514 - 3.806 - 9.518 \\ &= 54.19 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qitot}] &: \\ &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 67.514 + -11.051 + 9.518 \\ &= 65.98 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qotot}] &: \\ &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 67.514 - -11.051 - 9.518 \\ &= 69.05 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pitot}] &: \\ &= \sigma_{3pm} - \sigma_{3pb} \\ &= 10.033 - 26.031 \\ &= -16.00 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3potot}] &: \\ &= \sigma_{3pm} + \sigma_{3pb} \\ &= 10.033 + 26.031 \\ &= 36.06 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qitot}] &: \\ &= \sigma_{3qm} - \sigma_{3qb} \\ &= -29.132 - -75.581 \\ &= 46.45 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qotot}] &: \\ &= \sigma_{3qm} + \sigma_{3qb} \\ &= -29.132 + -75.581 \\ &= -104.71 \text{ N./mm}^2 \end{aligned}$$

Allowable stresses:

$$\begin{aligned} \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\ &= 164.0(3 - (1 / 1.5)) \cdot (71.351 / 164.0)^2 \\ &= 471.30 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\ &= 164.0(3 - (1 / 1.5)) \cdot (56.374 / 164.0)^2 \\ &= 479.08 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\ &= 164.0(3 - (1 / 1.5)) \cdot (67.514 / 164.0)^2 \\ &= 473.47 \text{ N./mm}^2 \end{aligned}$$



For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2) \\ &= 164.0(3 - (1/1.5) \cdot (67.514/164.0)^2) \\ &= 473.47 \text{ N./mm}^2 \end{aligned}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2) \\ &= 164.0(3 - (1/1.5) \cdot (10.033/164.0)^2) \\ &= 491.59 \text{ N./mm}^2 \end{aligned}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2) \\ &= 164.0(3 - (1/1.5) \cdot (-29.132/164.0)^2) \\ &= 488.55 \text{ N./mm}^2 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm2)

Stress	Actual	Allowed	Result	Equation
Stot1pi	77.06	471.30	Pass	46
Stot1po	65.64	471.30	Pass	47
Stot1qi	76.94	479.08	Pass	48
Stot1qo	35.81	479.08	Pass	49
Stot2pi	80.84	473.47	Pass	50
Stot2po	54.19	473.47	Pass	51
Stot2qi	65.98	473.47	Pass	52
Stot2qo	69.05	473.47	Pass	53
Stot3pi	-16.00	491.59	Pass	54
Stot3po	36.06	491.59	Pass	55
Stot3qi	46.45	488.55	Pass	56
Stot3qo	-104.71	488.55	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	10260.214 mm.
Center of Gravity of the Liquid	7049.999 mm.
Center of Gravity of the Nozzles	4259.371 mm.
Center of Gravity of the Added Weights (Operating)	11682.771 mm.
Center of Gravity of the Added Weights (Empty)	11682.771 mm.
Center of Gravity of Bare Shell New and Cold	6612.107 mm.
Center of Gravity of Bare Shell Corroded	6583.453 mm.
Vessel CG in the Operating Condition	9180.312 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	8426.313 mm.
Vessel CG in the Test Condition	7458.438 mm.

Rigging Analysis Results:

• Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	12100.00 mm.
Total vessel weight (No Liquid) Twt	70472.30 Kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	105708.46 Kgf
Elevation of the Tailing Lug (bottom)	200.00 mm.
Elevation of the Lifting Lug (top)	8800.00 mm.
Design Reaction force at the tailing lug	4593.24 Kgf
Design Reaction force at the lifting lug	101115.22 Kgf
CG distance during field lifting operation	8426.31 mm.
CG distance from tailing lug	8226.31 mm.
CG distance from the nearer lifting lug/trunnion	373.69 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm2, mm.)

	Max Stress	Elevation	Allowables
Bending	4.77	8850.00	99.81
Shear	3.76	0.00	122.50 (0.7*5)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm2	Shear Force Kgf	Shear Stress N./mm2
0.00	0.0	0.0	33821.7	3.8
3250.00	1432.3	0.1	24176.2	0.9
6850.00	56451.7	2.4	42044.7	1.6
9650.00	80425.3	3.4	19678.7	0.7
12050.00	12060.8	0.6	4303.5	0.2

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0478 at elevation 8850.0 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated -----





Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3516.0000	mm.
Skirt Thickness	STHK	8.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.5000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	25.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	175.00	N./mm2
Basing Yield Stress		262.00	N./mm2
Inside Diameter of Basing	DI, D4	3430.0000	mm.
Outside Diameter of Basing	DOU, D5	3760.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	30.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Root Area of a Single Bolt	Area	5.4000	cm2
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	137.30	N./mm2
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	3639.9998	mm.
Bolt Allowable Shear Stress		107.873	N./mm2
Ultimate Comp. Strength of Concrete	FPC	35.0	N./mm2
Allowable Comp. Strength of Concrete	FC	8.8	N./mm2
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	10.0000	mm.
Width of Gussets at Top Plate	TWDT	102.0000	mm.
Width of Gussets at Base Plate	BWDT	102.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm2
Height of Gussets	HG, h1	185.0000	mm.
Distance between Gussets	RG, b6	90.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	25.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	120.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	57.0000	mm.
External Corrosion Allowance	CA	0.0000	mm.
Dead Weight of Vessel	DW	70472.3	Kgf
Operating Weight of Vessel	ROW	72115.9	Kgf
Earthquake Moment on Basing	EQMOM	157357.9	Kg-m.
Wind Moment on Basing	WIMOM	155150.2	Kg-m.
User Moment at the Base [Earthquake Case]		46473.0	Kg-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		46473.0	Kg-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 203831/3640.0) - 72116) / 20$$

$$= 7593.5 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 201623/3640.0) - 70472) / 20$$

$$= 7554.4 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3640.0) - 72116) / 20$$

$$= -3605.8 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	5.4000	cm ²
Area Available in all the Bolts	Abss * RN :	108.0000	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.173	0.259	1.129	2.721	0.465	0.773	29072.9	101188.8	4010.2
4	0.216	0.238	1.268	2.626	0.456	0.777	29795.4	101911.4	4259.5
6	0.248	0.254	1.365	2.555	0.449	0.779	30355.6	102471.5	4460.1
8	0.251	0.251	1.373	2.549	0.448	0.779	30402.9	102518.9	4477.4

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 30402.9 / (0.944 \cdot 3640.0 \cdot 2.549)$$

$$= 68.06 \text{ N./mm}^2, \text{ Should be } \leq 137.3 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 5.4 / (\pi \cdot 3640.0)$$

$$= 0.944 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 2.321[(2 \cdot 0.2511 \cdot 3640.0 + 165.0) / (2 \cdot 0.2511 \cdot 3640.0)]$$

$$= 2.531 \text{ N./mm}^2, \text{ Should be } \leq 8.8 \text{ N./mm}^2$$

Values for table 10.3, l = 122.0, b = 285.885, l/b = 0.426745

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(418.715, 1409.436) = 1409.436 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 1409.436 / 172.9)} + 0.0$$

$$= 21.9 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 120.0, b = 90.0, b/l = 0.75

As b/l (0.75) is less than 1, inverting b/l = 1.333.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 7560.24 / (4 \cdot \pi) [1.3(\ln((2 \cdot 120.0 \cdot \sin(\pi \cdot 62.0/120.0)) / (\pi \cdot 25.4))) + 1] - [(0.7 - 0.091) \cdot 7560.24 / (4 \cdot \pi)]$$

$$= 1095.2416 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb)} + Ca$$



$$= \sqrt{6 \cdot \text{Abs}(1095.24/174.66)} + 0.0$$

$$= 19.2085 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 7560.24 / (124.11 \cdot 102.0) + 0.0$$

$$= 9.525 \text{ (not less than } 9.525 + 0.0 \text{) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3640.0/20 = 571.77 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76(7560 \cdot 62.0 / (571.77 \cdot 210.0 \cdot 246))^{2/3} \cdot 1758.01^{1/3} + \text{Ca}$$

$$= 6.146 + 1.5 = 7.646 \text{ mm.}$$

Shear Stress in a Single Bolt [rb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 16078.1 / (5.4 \cdot 20)$$

$$= 14.6 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	21.9000	mm.
Actual Basing Thickness as entered by user	25.0000	mm.
Required Thickness of Chair Cap	19.2085	mm.
Actual Top Ring Thickness as entered by user	25.0000	mm.
Required Gusset thickness, + CA	9.5250	mm.
Actual Gusset Thickness as entered by user	10.0000	mm.
Required Thickness of Skirt for Local Stress	7.6457	mm.
Given Thickness of Skirt	8.0000	mm.
Required Gusset Height to meet local stress	116.3978	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 43.625 \cdot (8.0 - 1.5)$$

$$= 28.91 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 29 / [(0.4 \cdot 246) \cdot 2 \cdot 0.707]$$

$$= 2.038 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 7560.2 / (170.0 + 2 \cdot (185.0 + 25.0))$$

$$= 12.814 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 7560.2 \cdot 62.0 / (170.0 \cdot (210.0) + 0.6667 \cdot (210.0)^2)$$

$$= 7.2 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{12.81^2 + 7.2^2}$$

$$= 14.698 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 14.7 / [(0.4 \cdot 246) \cdot 2 \cdot 0.707]$$

$$= 1.036 \text{ mm.}$$



Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= 7560.2/(2 • 102.0)
= 37.06 Kgf/mm.

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= 7560.2 • 62.0/(2 • 210.0 • 102.0)
= 10.942 Kgf/mm.

Resultant Weld Load [Wr]:
= $\sqrt{Wv^2 + Wh^2}$
= $\sqrt{37.06^2 + 10.94^2}$
= 38.641 Kgf/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 38.64/[(0.4 • 246) • 2 • 0.707]
= 2.724 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	2.7236	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N33"

From: 20

Pressure for Reinforcement Calculations	P	16.599	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm2
Parent Allowable Stress At Ambient	Sva	175.00	N./mm2
Inside Diameter of Elliptical Head	D	3500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm.
Head Internal Corrosion Allowance	c	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	151.00	N./mm2
Allowable Stress At Ambient	Sna	161.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm2
Pad Allowable Stress At Ambient	Spa	175.00	N./mm2
Diameter of Pad along vessel surface	Dp	178.9000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		45.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		270.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2000.0	mm.
Centerline Distance to Tangent		1200.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N33"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14



Actual Inside Diameter Used in Calculation 2.734 in.
Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
= $0.5 \cdot 75.431 \cdot (\exp(16.599/(151.0 \cdot 1.0)) - 1) + 3.0 + 0.0$
= $0.4169 + 3.0 + 0.0 = 3.4169$ mm. Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
= $S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
= $151.0 \cdot 1.0 \cdot \ln((2 \cdot 6.735 + 75.431) / 75.431) - 0.0$
= $248.071 - 0.0 = 248.0712$ bars

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.800 mm
Min. required shell thickness, internal p + c : 25.596 mm.

Nozzle neck minimum required thickness : 7.800 mm.

Minimum nozzle neck thickness 9.728 mm., is greater than required thickness 7.800 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.7	Allowable	: 151.0 N./mm ²	Passed
Expansion	: 0.0	Allowable	: 372.2 N./mm ²	Passed
Occasional	: 4.3	Allowable	: 200.8 N./mm ²	Passed
Shear	: 9.4	Allowable	: 105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
= $\min[S_n/S, 1]$
= $\min[151.0/164.0, 1]$
= 0.921

Pad Material Factor [frp]:
= $\min[S_p/S, 1]$
= $\min[164.0/164.0, 1]$
= 1.000

Thickness of Nozzle [tn]:
= thickness - corrosion allowance
= $9.735 - 3.0$
= 6.735 mm.

Effective Pressure Radius [Reff]:
= $0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
= $0.9/6 \cdot 3506.0 [2 + (3506.0/(2 \cdot 878.0))^2]$
= 3148.218 mm.

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, L_r is $\min[LR1, LR2, LR3]$:

Limit [LR1]:
= $\sqrt{(Reff \cdot t) + W}$
= $\sqrt{(3148.218 \cdot 17.0) + 45.0}$
= 276.343 mm.

Limit [LR2]:



$$\begin{aligned} &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\ &= \sqrt{((3148.218 + 17.0) \cdot (17.0 + 20.0))} \\ &= 342.218 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Limit [LR3]:} \\ &= 2 \cdot R_n \\ &= 2 \cdot 37.715 \\ &= 75.431 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Final value of [LR]:} \\ &= \min(LR1, LR2, LR3) \\ &= \min(276.343, 342.218, 75.431) \\ &= 75.431 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Intermediate value [Cn]:} \\ &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min((17.0 + 20.0)/6.735)^{0.35}, 1) \\ &= \min(1.815, 1) \\ &= 1.000 \end{aligned}$$

$$\begin{aligned} \text{Intermediate value [Cp]:} \\ &= \exp((0.35 \cdot d_i - X_o)/(16t)) \\ &= \exp((0.35 \cdot 3506.0 - 44.45)/(16 \cdot 17.0)) \\ &= 77.322 \end{aligned}$$

$$\begin{aligned} \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\ &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\ &= \min(17.0 + 20.0 + 1.0 \cdot \sqrt{(37.715 \cdot 6.735)}, 200.0 + 17.0) \\ &= 52.937 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Effective Vessel Thickness [teff]:} \\ &= t + ((A_5 \cdot frp) / LR) \\ &= 17.0 + ((9.0 \cdot 1.0) / 75.431) \\ &= 28.931 \text{ mm.} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned} \text{Area Contributed by the Vessel Wall [A1]:} \\ &= t \cdot LR \\ &= 17.0 \cdot 75.431 \\ &= 12.823 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\ &= t_n \cdot LH \\ &= 6.735 \cdot 52.937 \\ &= 3.565 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Pad Fillet Weld [A42]:} \\ &= 0.5 \cdot Leg42^2 \\ &= 0.5 \cdot 20.0^2 \\ &= 2.000 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Outside Fillet Weld [A41]:} \\ &= 0.5 \cdot Leg41^2 - \text{Area cut by thickness limit} \\ &= 0.5 \cdot 20.0^2 - 0.083 \\ &= 1.917 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Reinforcing Pad [A5]:} \\ &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(45.0 \cdot 20.0, 75.431 \cdot 20.0) \\ &= 9.000 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{The total area contributed by A1 through A5 [AT]:} \\ &= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5) \\ &= 12.823 + 0.921(3.565 + 0.0) + 1.917 + 2.0 + 0.0 + 1.0(9.0) \\ &= 29.023 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Nozzle Radius for Force Calculation [Rxn]:} \\ &= t_n / \ln[1 + t_n/R_n] \\ &= 6.735 / \ln[1 + 6.735/37.715] \end{aligned}$$



= 40.991 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $28.931 / \ln[1 + 28.931/3148.218]$
= 3162.682 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S \cdot E$
= $1.5 \cdot 164.0 \cdot 1.0$
= 246.0 N./mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $16.599 \cdot 40.991 \cdot 52.937$
= 367.3 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} \cdot (L_R + t_n)/2$
= $16.599 \cdot 3162.682 \cdot (75.431 + 6.735)/2$
= 21993.8 Kgf

Discontinuity Force from Internal Pressure [fY]:

= $(P \cdot R_{xs} \cdot R_{nc})/2$
= $(16.599 \cdot 3162.682 \cdot 37.715)/2$
= 10095.6 Kgf

Area Resisting Internal Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(21993.8 + 10095.6 + 367.3)/16.599$
= 1917.4 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$
= $246.0 / (2 \cdot 1917.425/29.023 - 3162.682/(2 \cdot 28.931))$
= 31.8 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $2 \cdot S[t/R_{xs}]$
= $2 \cdot 164.0[17.0/3162.682]$
= 17.6 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(31.751, 17.63)$
= 17.630 bars

Average Primary Membrane Stress [σ_{Avg}]:

= $(f_N + f_S + f_Y) / A_T$
= $(367.309 + 21993.811 + 10095.562)/29.023$
= 109.670 N./mm²

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot R_{xs} / (2 \cdot t_{eff})$
= $16.599 \cdot 3162.682 / (2 \cdot 28.931)$
= 90.7 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 109.67 - 90.734, 90.734)$
= 128.6 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm ²
Local Primary Membrane Stress	PL	128.61	N./mm ²
Maximum Allowable Working Pressure	Pmax	17.63	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:



$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (37.715 + 6.735) / 37.715 \\ &= 1.179 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (37.715 + 6.735) \\ &= 69.822 \text{ mm.} \end{aligned}$$

Weld Length of Pad to Shell Weld [L_{TP}]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (37.715 + 6.735 + 45.0) \\ &= 140.508 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 14.142, 14.142, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

tc, L_{41T} = 14.142, must be $\geq \min(0.7 \cdot 6.735, 6.000)$ Passed

tf1, L_{42T} = 14.142, must be $\geq \min(0.6 \cdot 20.000, 0.6 \cdot 17.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(10096 \cdot 1.18, 1.5 \cdot 151.0(3.565 + 0.0), \pi/4 \cdot 16.6 \cdot 37.72^2 \cdot 1.18^2) \\ &= 262.675 \text{ Kg} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 262.7 \cdot 17.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164) \\ &= 120.689 \text{ Kg} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 262.7 \cdot 20.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164) \\ &= 141.987 \text{ Kg} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot tw_1 + 0.49 \cdot L_{43T})) \\ &= 120.689 / (69.822 \cdot (0.6 \cdot 17.0 + 0.49 \cdot 0.0)) \\ &= 1.662 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot tw_2 + 0.49 \cdot L_{41T})) \\ &= 141.987 / (69.822 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 14.142)) \\ &= 1.054 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T})) \\ &= 141.987 / (140.508 \cdot (0.49 \cdot 14.142)) \\ &= 1.430 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(1.662, 1.054, 1.43) \\ &= 1.7 \text{ must be less than or equal to } 164.0 \text{ N./mm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is : -104 °C

The computed stress ratio is : 0.058

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$



$$= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0)$$
$$= 0.058$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.058

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0)$
 $= 0.058$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.058

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0)$
 $= 0.058$

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : 4 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 9 °C
Temperature reduction per Graph 3.12 : 5 °C
The computed stress ratio is : 0.882

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (14.99 \cdot 1.0) / (20.0 - 3.0 - 0.0)$
 $= 0.882$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.058

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0)$
 $= 0.058$

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : 4 °C
Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -26 °C



The Drop for this Nozzle is : 0.3120 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.3119 mm.

Input Echo, WRC107/537 Item 1, Description: N33" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	6300.000	mm.
Vessel Thickness	Tv	20.000	mm.
Design Temperature	T1	65.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm2
Vessel Hot S.I. Allowable	Smh	164.00	N./mm2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	69.431	mm.
Nozzle Thickness	Tn	9.735	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	161.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	151.00	N./mm2
Thickness of Reinforcing Pad	Tpad	20.000	mm.
Diameter of Reinforcing Pad	Dpad	178.900	mm.
Design Internal Pressure	Dp	16.599	bars
Include Pressure Thrust		No	

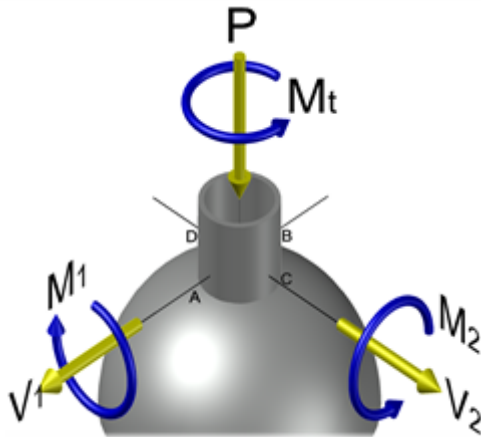
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	81.6	Kgf
Longitudinal Shear (SUS)	(V1) V1	122.4	Kgf
Circumferential Shear (SUS)	(Vc) V2	122.4	Kgf
Circumferential Moment (SUS)	(Mc) M1	24.5	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	36.7	Kg-m.
Torsional Moment (SUS)	Mt	44.9	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(3161.5 (20.0 - 3.0))}$$

$$= 853.941 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	81.6	Kgf
Circumferential Shear	(VC) V2	122.4	Kgf
Longitudinal Shear	(VL) V1	122.4	Kgf
Circumferential Moment	(MC) M1	24.5	Kg-m.
Longitudinal Moment	(ML) M2	36.7	Kg-m.
Torsional Moment	MT	44.9	Kg-m.

Dimensionless Param: U = 0.13 TAU = 6.10 RHO = 4.00 (5.49)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.10111	(A,B,C,D)
$M(x) / P$	SP 4	0.03922	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.42034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.18444	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.42034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.18444	(A,B,C,D)
$N(y) * T / P$	SP 4	0.33643	(A,B,C,D)
$M(y) / P$	SP 4	0.27446	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10306	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.41541	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10306	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.41541	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad. Bend.	P	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-0.6	0.6	0.6	-0.6
Rad. Memb.	ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-0.8	0.8	0.8	-0.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.4	0.6	1.0	-0.4	-1.0	0.4	0.6	-0.3



Tang. Memb. P	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Tang. Bend. P	-1.0	1.0	-1.0	1.0	-1.0	1.0	-1.0	1.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-4.3	4.3	4.3	-4.3
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.5	6.5	6.5	-6.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-7.8	7.2	5.4	-5.7	-5.6	5.1	3.2	-3.5
Shear VC	0.2	0.2	-0.2	-0.2	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.2	1.2	0.7	0.7	0.7	0.7	1.2	1.2
Str. Int.	8.0	7.4	5.6	5.8	5.7	5.2	3.7	4.0

Unitless Prm: U = 0.39 TAU = 0.00 (12.78) RHO = 0.00 (2.52)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.16376	(A,B,C,D)
M(x) / P	SR 2	0.11042	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.15810	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.34695	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.15810	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.34695	(A,B,C,D)
N(y) * T / P	SR 2	0.04917	(A,B,C,D)
M(y) / P	SR 2	0.03311	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04838	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.10445	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04838	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.10445	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Rad. Bend. P		-1.8	1.8	-1.8	1.8	-1.8	1.8	-1.8	1.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-7.5	7.5	7.5	-7.5
Rad. Memb. ML		-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-11.2	11.2	11.2	-11.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-14.3	11.7	9.7	-9.0	-10.3	8.3	5.7	-5.5
Tang. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Tang. Bend. P		-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-2.2	2.2	2.2	-2.2
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-3.4	3.4	3.4	-3.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.3	3.5	2.9	-2.7	-3.1	2.5	1.7	-1.7
Shear VC		0.3	0.3	-0.3	-0.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tot. Shear		0.8	0.8	0.3	0.3	0.3	0.3	0.8	0.8



Str. Int.	14.4	11.8	9.8	9.0	10.3	8.3	5.9	5.7
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WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad. Pl (SUS)		-0.4	-0.4	0.3	0.3	-0.3	-0.3	0.2	0.2
Rad. Q (SUS)		-1.0	1.0	0.7	-0.7	-0.7	0.7	0.4	-0.4
Long. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long. Pl (SUS)		-0.3	-0.3	-0.1	-0.1	-0.2	-0.2	-0.1	-0.1
Long. Q (SUS)		-7.5	7.5	5.6	-5.6	-5.3	5.3	3.4	-3.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.2	0.2	-0.2	-0.2	-0.2	-0.2	0.2	0.2
Shear Q (SUS)		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl (SUS)		71.1	71.1	71.5	71.5	71.1	71.1	71.4	71.4
Pm+Pl+Q (Total)		70.0	78.6	76.7	70.8	70.3	76.3	74.8	71.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.52	246.00	Passed
Pm+Pl+Q (TOTAL)	78.56	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad. Pl (SUS)		-1.3	-1.3	0.4	0.4	-1.0	-1.0	0.1	0.1
Rad. Q (SUS)		-13.0	13.0	9.4	-9.4	-9.3	9.3	5.6	-5.6
Long. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long. Pl (SUS)		-0.4	-0.4	0.1	0.1	-0.3	-0.3	0.0	0.0
Long. Q (SUS)		-3.9	3.9	2.8	-2.8	-2.8	2.8	1.7	-1.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.3	0.3	-0.3	-0.3	-0.3	-0.3	0.3	0.3
Shear Q (SUS)		0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl (SUS)		70.8	70.8	71.7	71.7	70.9	70.9	71.5	71.5
Pm+Pl+Q (Total)		66.9	82.9	80.9	68.5	68.1	79.4	77.0	69.6



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.69	246.00	Passed
Pm+Pl+Q (TOTAL)	82.93	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: MH124"

From: 30

Pressure for Reinforcement Calculations	P	16.599	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm2
Parent Allowable Stress At Ambient	Sva	175.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3500.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1050.00	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	164.00	N./mm2
Allowable Stress At Ambient	Sna	175.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm2
Pad Allowable Stress At Ambient	Spa	175.00	N./mm2
Diameter of Pad along vessel surface	Dp	1119.6000	mm.
Thickness of Pad	te	24.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm.
Reinforcing Pad Width		255.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH124"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.798	in.
Actual Thickness Used in Calculation	0.601	in.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 585.063 \cdot (\exp(16.599/(164.0 \cdot 1.0)) - 1) + 3.0 + 0.0 \\ &= 2.976 + 3.0 + 0.0 = 5.976 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 164.0 \cdot 1.0 \cdot \ln((2 \cdot 12.269 + 585.063) / 585.063) - 0.0 \\ &= 67.373 - 0.0 = 67.3734 \text{ bars} \end{aligned}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[164.0/164.0, 1] \\ &= 1.000 \end{aligned}$$

Pad Material Factor [frp]:

$$\begin{aligned} &= \min[S_p/S, 1] \\ &= \min[164.0/164.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 15.269 - 3.0 \\ &= 12.269 \text{ mm.} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 3506.0/21.0 \\ &= 166.952 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3500.0/2 + 3.0 \\ &= 1753.000 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned} &= \sqrt{(Reff \cdot t) + W} \\ &= \sqrt{(1753.0 \cdot 21.0) + 255.0} \\ &= 446.867 \text{ mm.} \end{aligned}$$

Limit [LR2]:

$$\begin{aligned} &= \sqrt{((Reff + t) \cdot (t + t_e))} \\ &= \sqrt{((1753.0 + 21.0) \cdot (21.0 + 24.0))} \\ &= 282.542 \text{ mm.} \end{aligned}$$

Limit [LR3]:

$$\begin{aligned} &= 2 \cdot R_n \\ &= 2 \cdot 292.531 \\ &= 585.063 \text{ mm.} \end{aligned}$$

Final value of [LR]:

$$\begin{aligned} &= \min(LR1, LR2, LR3) \\ &= \min(446.867, 282.542, 585.063) \\ &= 282.542 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 21.0, 24.0] + \sqrt{(292.531 \cdot 12.269)} \\ &= 83.908 \text{ mm.} \end{aligned}$$



Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(21.0 + 24.0)$$

$$= 360.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[83.908, 250.0, 360.0] + 21.0$$

$$= 104.908 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 21.0 + ((61.2 \cdot 1.0) / 282.542)$$

$$= 42.660 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{ (D_i + t_{eff}) \cdot t_{eff} }))$$

$$= \min(12, (2 \cdot 292.53 + 12.269) / (\sqrt{ (3506.0 + 42.66) \cdot 42.66 }))$$

$$= 1.535$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 21.0 \cdot 282.542 \cdot \max((1.535/5)^{0.85}, 1)$$

$$= 59.334 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 12.269 \cdot 104.908$$

$$= 12.871 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot Leg_{42}^2$$

$$= 0.5 \cdot 20.0^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg_{41}^2$$

$$= 0.5 \cdot 20.0^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(255.0 \cdot 24.0, 282.542 \cdot 24.0)$$

$$= 61.200 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5)$$

$$= 59.334 + 1.0(12.871 + 0.0) + 2.0 + 2.0 + 0.0 + 1.0(61.2)$$

$$= 137.405 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 12.269 / \ln[1 + 12.269 / 292.531]$$

$$= 298.623 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff} / R_{eff}]$$

$$= 42.66 / \ln[1 + 42.66 / 1753.0]$$

$$= 1774.242 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 164.0 \cdot 1.0$$



$$= 246.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 16.599 \cdot 298.623 \cdot 104.908 \\ &= 5302.9 \text{ Kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 16.599 \cdot 1774.242 (282.542 + 12.269) \\ &= 88539.2 \text{ Kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 16.599 \cdot 1774.242 \cdot 292.531 \\ &= 87854.8 \text{ Kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (88539.2 + 87854.8 + 5302.9) / 16.599 \\ &= 10734.2 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 246.0 / (2 \cdot 10734.151 / 137.405 - 1774.242 / 42.66) \\ &= 21.5 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / R_{xs} \\ &= 164.0 [21.0 / 1774.242] \\ &= 19.4 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(21.455, 19.41) \\ &= 19.410 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (5302.875 + 88539.242 + 87854.75) / 137.405 \\ &= 129.681 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 16.599 \cdot 1774.242 / 42.66 \\ &= 69.0 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 129.681 - 69.039, 69.039) \\ &= 190.3 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm ²
Local Primary Membrane Stress	PL	190.32	N./mm ²
Maximum Allowable Working Pressure	Pmax	19.41	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (292.531 + 12.269) / 292.531 \\ &= 1.042 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (R_n + t_n) \\ &= \pi / 2 \cdot (292.531 + 12.269) \\ &= 478.779 \text{ mm.} \end{aligned}$$

Weld Length of Pad to Shell Weld [LtP]:



$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (292.531 + 12.269 + 255.0) \\ &= 879.332 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 14.142, 14.142, 0.000, mm.

Required Weld Throat Dimensions :
tc, L41T = 14.142, must be $\geq \min(0.7 \cdot 12.269, 6.000)$ Passed
tf1, L42T = 14.142, must be $\geq \min(0.6 \cdot 24.000, 0.6 \cdot 21.000)$ Passed

Weld Load Value [fwelds]:
= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(87855 \cdot 1.04, 1.5 \cdot 164.0(12.871 + 0.0), \pi/4 \cdot 16.6 \cdot 292.53^2 \cdot 1.04^2)$
= 12350.929 Kgf

Discontinuity Force [fws]:
= $fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$
= $12350.9 \cdot 21.0 \cdot 164 / (21.0 \cdot 164 + 24.0 \cdot 164)$
= 5763.766 Kgf

Discontinuity Force [fwp]:
= $fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$
= $12350.9 \cdot 24.0 \cdot 164 / (21.0 \cdot 164 + 24.0 \cdot 164)$
= 6587.162 Kgf

Shear Stress [τ_1]:
= $fws / (L_T \cdot (0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
= $5763.766 / (478.779 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 0.0))$
= 9.838 N./mm²

Shear Stress [τ_2]:
= $fwp / (L_T \cdot (0.6 \cdot tw_2 + 0.49 \cdot L_{41T}))$
= $6587.162 / (478.779 \cdot (0.6 \cdot 24.0 + 0.49 \cdot 14.142))$
= 6.326 N./mm²

Shear Stress [τ_3]:
= $fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$
= $6587.162 / (879.332 \cdot (0.49 \cdot 14.142))$
= 10.601 N./mm²

Maximum Shear Stress in the Welds:
= $\max(\tau_1, \tau_2, \tau_3)$
= $\max(9.838, 6.326, 10.601)$
= 10.6 must be less than or equal to 164.0 N./mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.227

The computed thickness ratio [sr]:
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(2.788 \cdot 1.0) / (15.269 - 3.0 - 0.0)$
= 0.227

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B



As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.227

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.788 • 1.0)/(15.269 - 3.0 - 0.0)
= 0.227

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.227

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.788 • 1.0)/(15.269 - 3.0 - 0.0)
= 0.227

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 4 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 13 °C
Temperature reduction per Graph 3.12 : 8 °C
The computed stress ratio is : 0.796

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (16.708 • 1.0)/(24.0 - 3.0 - 0.0)
= 0.796

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.227

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.788 • 1.0)/(15.269 - 3.0 - 0.0)
= 0.227

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : 4 °C
Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 26.7482 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 300.7483 mm.



Input, Nozzle Desc: N14"

From: 30

Pressure for Reinforcement Calculations	P	16.599	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm2
Parent Allowable Stress At Ambient	Sva	175.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3500.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		400.00	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	151.00	N./mm2
Allowable Stress At Ambient	Sna	161.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm2
Pad Allowable Stress At Ambient	Spa	175.00	N./mm2
Diameter of Pad along vessel surface	Dp	204.3000	mm.
Thickness of Pad	te	24.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm.
Reinforcing Pad Width		45.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N14"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	3.624	in.
Actual Thickness Used in Calculation	0.438	in.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 98.05 \cdot (\exp(16.599/(151.0 \cdot 1.0)) - 1) + 3.0 + 0.0 \\ &= 0.5419 + 3.0 + 0.0 = 3.5419 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq} \\ &= 151.0 \cdot 1.0 \cdot \ln((2 \cdot 8.125 + 98.05)/98.05) - 0.0 \\ &= 231.55 - 0.0 = 231.5496 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 8.270 mm
Min. required shell thickness, internal p + c : 20.834 mm.

Nozzle neck minimum required thickness : 8.270 mm.

Minimum nozzle neck thickness 9.728 mm., is greater than required thickness 8.270 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 20.7,	Allowable	: 151.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 369.3 N./mm ²	Passed
Occasional	: 5.7,	Allowable	: 200.8 N./mm ²	Passed
Shear	: 10.0,	Allowable	: 105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[151.0/164.0, 1] \\ &= 0.921 \end{aligned}$$

Pad Material Factor [frp]:

$$\begin{aligned} &= \min[S_p/S, 1] \\ &= \min[164.0/164.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 11.125 - 3.0 \\ &= 8.125 \text{ mm.} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 3506.0/21.0 \\ &= 166.952 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3500.0/2 + 3.0 \\ &= 1753.000 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned} &= \sqrt{(Reff \cdot t)} + W \\ &= \sqrt{(1753.0 \cdot 21.0)} + 45.0 \\ &= 236.867 \text{ mm.} \end{aligned}$$

Limit [LR2]:

$$= \sqrt{((Reff + t) \cdot (t + t_e))}$$



$$= \sqrt{((1753.0 + 21.0) \cdot (21.0 + 24.0))}$$
$$= 282.542 \text{ mm.}$$

Limit [LR3]:

$$= 2 \cdot R_n$$
$$= 2 \cdot 49.025$$
$$= 98.050 \text{ mm.}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$
$$= \min(236.867, 282.542, 98.05)$$
$$= 98.050 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$
$$= \min[1.5 \cdot 21.0, 24.0] + \sqrt{(49.025 \cdot 8.125)}$$
$$= 43.958 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 200.0$$
$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(21.0 + 24.0)$$
$$= 360.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[43.958, 200.0, 360.0] + 21.0$$
$$= 64.958 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot f_{rp}) / LR)$$
$$= 21.0 + ((10.8 \cdot 1.0) / 98.05)$$
$$= 32.015 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 49.02 + 8.125) / (\sqrt{(3506.0 + 32.015) \cdot 32.015}))$$
$$= 0.315$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 21.0 \cdot 98.05 \cdot \max((0.315/5)^{0.85}, 1)$$
$$= 20.590 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$
$$= 8.125 \cdot 64.958$$
$$= 5.278 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}42^2$$
$$= 0.5 \cdot 20.0^2$$
$$= 2.000 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}41^2$$
$$= 0.5 \cdot 20.0^2$$
$$= 2.000 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$
$$= \min(45.0 \cdot 24.0, 98.05 \cdot 24.0)$$
$$= 10.800 \text{ cm}^2$$



The total area contributed by A1 through A5 [AT]:
= $A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$
= $20.59 + 0.921(5.278 + 0.0) + 2.0 + 2.0 + 0.0 + 1.0(10.8)$
= 40.250 cm^2

Nozzle Radius for Force Calculation [Rxn]:
= $t_n / \ln[1 + t_n/R_n]$
= $8.125 / \ln[1 + 8.125/49.025]$
= 52.984 mm .

Shell Radius for Force Calculation [Rxs]:
= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $32.015 / \ln[1 + 32.015/1753.0]$
= 1768.961 mm .

Allowable Local Primary Membrane Stress [Sallow]:
= $1.5 \cdot S \cdot E$
= $1.5 \cdot 164.0 \cdot 1.0$
= 246.0 N./mm^2

Force acting on the Nozzle [fN]:
= $P \cdot R_{xn} \cdot L_H$
= $16.599 \cdot 52.984 \cdot 64.958$
= 582.6 Kg

Force acting on the Shell [fS]:
= $P \cdot R_{xs} (L_R + t_n)$
= $16.599 \cdot 1768.961 (98.05 + 8.125)$
= 31792.3 Kg

Discontinuity Force from Internal Pressure [fY]:
= $P \cdot R_{xs} \cdot R_{nc}$
= $16.599 \cdot 1768.961 \cdot 49.025$
= 14679.7 Kg

Area Resisting Internal Pressure [Ap]:
= $(f_S + f_Y + f_N) / P$
= $(31792.3 + 14679.7 + 582.6) / 16.599$
= 2779.8 cm^2

Maximum Allowable Working Pressure Candidate [Pmax1]:
= $S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$
= $246.0 / (2 \cdot 2779.837 / 40.25 - 1768.961 / 32.015)$
= 29.7 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:
= $S / [t / R_{xs}]$
= $164.0 [21.0 / 1768.961]$
= 19.5 bars

Maximum Allowable Working Pressure [Pmax]:
= $\min(P_{max1}, P_{max2})$
= $\min(29.682, 19.468)$
= 19.468 bars

Average Primary Membrane Stress [σ_{Avg}]:
= $(f_N + f_S + f_Y) / AT$
= $(582.584 + 31792.26 + 14679.654) / 40.25$
= 114.647 N./mm^2

General Primary Membrane Stress [σ_{Circ}]:
= $P \cdot R_{xs} / t_{eff}$
= $16.599 \cdot 1768.961 / 32.015$
= 91.7 N./mm^2

Maximum Local Primary Membrane Stress [PL]:
= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 114.647 - 91.723, 91.723)$
= 137.6 N./mm^2



Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm2
Local Primary Membrane Stress	PL	137.57	N./mm2
Maximum Allowable Working Pressure	Pmax	19.47	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (49.025 + 8.125) / 49.025 \\ &= 1.166 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (49.025 + 8.125) \\ &= 89.771 \text{ mm.} \end{aligned}$$

Weld Length of Pad to Shell Weld [LtP]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (49.025 + 8.125 + 45.0) \\ &= 160.457 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 14.142, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 14.142, \text{ must be } \geq \min(0.7 \cdot 8.125, 6.000) \text{ Passed} \\ t_{f1}, L42T &= 14.142, \text{ must be } \geq \min(0.6 \cdot 24.000, 0.6 \cdot 21.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(14680 \cdot 1.17, 1.5 \cdot 151.0(5.278 + 0.0), \pi/4 \cdot 16.6 \cdot 49.02^2 \cdot 1.17^2) \\ &= 434.214 \text{ Kg} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 434.2 \cdot 21.0 \cdot 164 / (21.0 \cdot 164 + 24.0 \cdot 164) \\ &= 202.633 \text{ Kg} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 434.2 \cdot 24.0 \cdot 164 / (21.0 \cdot 164 + 24.0 \cdot 164) \\ &= 231.581 \text{ Kg} \end{aligned}$$

Shear Stress [τ1]:

$$\begin{aligned} &= fws / (L_t \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 202.633 / (89.771 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 0.0)) \\ &= 1.845 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ2]:

$$\begin{aligned} &= fwp / (L_t \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\ &= 231.581 / (89.771 \cdot (0.6 \cdot 24.0 + 0.49 \cdot 14.142)) \\ &= 1.186 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ3]:

$$\begin{aligned} &= fwp / (L_{tP} \cdot (0.49 \cdot L_{42T})) \\ &= 231.581 / (160.457 \cdot (0.49 \cdot 14.142)) \\ &= 2.043 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(1.845, 1.186, 2.043) \\ &= 2.0 \text{ must be less than or equal to } 164.0 \text{ N./mm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C



User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.075

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.508 • 1.0)/(9.735 - 3.0 - 0.0)
= 0.075

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.075

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.508 • 1.0)/(9.735 - 3.0 - 0.0)
= 0.075

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.075

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.508 • 1.0)/(9.735 - 3.0 - 0.0)
= 0.075

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 4 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 13 °C
Temperature reduction per Graph 3.12 : 8 °C
The computed stress ratio is : 0.796

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (16.708 • 1.0)/(24.0 - 3.0 - 0.0)
= 0.796

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.075

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.508 • 1.0)/(9.735 - 3.0 - 0.0)
= 0.075



Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : 4 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -104 °C

The Drop for this Nozzle is : 0.9335 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.9335 mm.

Input Echo, WRC297 Item 1, Description: N14" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	3500.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 164.00	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	92.050	mm.
Nozzle Thickness	11.1252	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 161.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 151.00	N./mm ²
Diameter of Reinforcing Pad	204.300	mm.
Thickness of Reinforcing Pad	24.0000	mm.

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	16.60	bars
Radial Load	P	110.14	Kgf
Circumferential Shear	Vc	163.16	Kgf
Longitudinal Shear	Vl	163.16	Kgf
Circumferential Moment	Mc	48.94	Kg-m.
Longitudinal Moment	Ml	65.26	Kg-m.
Torsional Moment	Mt	81.57	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (4.762) is < 5
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1775.5(24.0 - 3.0))}$$

$$= 751.512 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3551.000 mm.
Nozzle Outside Diameter	(d) =	114.300 mm.
Vessel Thickness used	(T) =	45.000 mm.
Nozzle Thickness used	(t) =	8.125 mm.
T / t	=	5.538
d / t	=	14.067

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.286$$

Nr/P	=	0.167
Mr/P	=	0.106
M0/P	=	0.278
N0/P	=	0.292
MrD/Mc	=	0.108
NrDL/Mc	=	0.141
M0d/Mc	=	0.305
N0DL/Mc	=	-0.134
MrD/M1	=	0.108
NrDL/M1	=	0.144
M0D/M1	=	0.297
N0DL/M1	=	-0.096

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	63	65	63	65
Outplane Stress Summary	58	71	67	61

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	31	31	31	31
Inplane Stress Summary	29	33	33	30

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	58	71	67	61

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0



Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31

Outplane Stress Summary	27	36	34	28
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	65	63	65

Inplane Stress Summary	61	66	65	64
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	61	66	65	64
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	13	11	13

Hoop Stress Summary	11	13	10	12
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-2	2	-2	2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-18	18	18	-18
Axial Pressure Stress	5	5	5	5

Axial Stress Summary -24 16 30 -1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	5	5	3	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	38	20	31	15

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	13	11	13
Hoop Stress Summary	11	13	10	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-2	2	-2	2
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-18	14	24	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4
Shear Stress Summary	3	3	5	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	30	17	26	17

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1763.5(24.0 - 3.0))}$$



= 749.355 mm.

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3527.000 mm.
 Nozzle Outside Diameter (d) = 204.300 mm.
 Vessel Thickness used (T) = 21.000 mm.
 Nozzle Thickness used (t) = 53.125 mm.
 T / t = 0.395
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.751$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.120
 Mr/P = 0.266
 M0/P = 0.078
 N0/P = 0.144
 MrD/Mc = 0.312
 NrDL/Mc = 0.076
 M0d/Mc = 0.097
 N0DL/Mc = 0.026
 MrD/M1 = 0.231
 NrDL/M1 = 0.089
 M0D/M1 = 0.074
 N0DL/M1 = 0.094

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	137	139	137	139
Outplane Stress Summary	132	142	140	137

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-3	3	-3	3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-9	9	9	-9
Inplane Pressure Stress	68	68	68	68
Inplane Stress Summary	54	81	75	63

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	132	142	140	137

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	68	68	68	68
Outplane Stress Summary	64	72	70	66

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-3	3	-3	3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	137	139	137	139
Inplane Stress Summary	123	152	143	133

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	123	152	143	133

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	65	63	65	63	65	63	65
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-4	4	4	-4	-1	1	1	-1
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-1	1	1	-1	-3	3	3	-3

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	63.0	65.0	63.0	65.0	63.0	65.0	63.0	65.0
Pm+P1 (SUS)	63.0	65.0	63.0	65.0	63.0	65.0	63.0	65.0
Pm+P1+Q (Total)	59.0	69.0	67.0	61.0	62.0	66.0	64.0	64.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	164.00	Passed
Pm+P1 (SUS)	65.00	246.00	Passed
Pm+P1+Q (TOTAL)	69.00	492.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	137	139	137	139	137	139	137	139
Circ. P1 (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-4	4	2	-2	-12	12	6	-6
Long. Pm (SUS)	68	68	68	68	68	68	68	68
Long. P1 (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-12	12	6	-6	-4	4	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	137.0	139.0	137.0	139.0	137.0	139.0	137.0	139.0
Pm+P1 (SUS)	137.0	139.0	137.0	139.0	137.0	139.0	137.0	139.0
Pm+P1+Q (Total)	133.0	143.0	139.0	137.0	125.0	151.0	143.0	133.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	139.00	164.00	Passed
Pm+P1 (SUS)	139.00	246.00	Passed
Pm+P1+Q (TOTAL)	151.00	492.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	13	11	13	11	13	11	13
Circ. P1 (SUS)	0	0	0	0	0	0	0	0



Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-9	-9	9	9	-7	-7	7	7
Long. Q (SUS)	-20	20	16	-16	-16	16	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	11.0	13.0	11.0	13.0	11.0	13.0	11.0	13.0
Pm+Pl (SUS)	15.1	17.1	14.3	14.6	13.2	15.1	12.6	13.6
Pm+Pl+Q (Total)	36.4	19.7	30.5	16.2	29.6	16.5	25.7	16.4

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	151.00	Passed
Pm+Pl (SUS)	17.12	226.50	Passed
Pm+Pl+Q (TOTAL)	36.40	453.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N43"

From: 70

Pressure for Reinforcement Calculations	P	16.598	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm2
Parent Allowable Stress At Ambient	Sva	175.00	N./mm2
Inside Diameter of Elliptical Head	D	3500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm.
Head Internal Corrosion Allowance	c	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	151.00	N./mm2
Allowable Stress At Ambient	Sna	161.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm2
Pad Allowable Stress At Ambient	Spa	175.00	N./mm2
Diameter of Pad along vessel surface	Dp	166.2000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		38.6500	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N43"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.734	in.
Actual Thickness Used in Calculation	0.383	in.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 75.431 \cdot (\exp(16.598/(151.0 \cdot 1.0)) - 1) + 3.0 + 0.0 \\ &= 0.4169 + 3.0 + 0.0 = 3.4169 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 151.0 \cdot 1.0 \cdot \ln((2 \cdot 6.735 + 75.431) / 75.431) - 0.0 \\ &= 248.071 - 0.0 = 248.0712 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.800 mm
Min. required shell thickness, internal p + c : 25.594 mm.

Nozzle neck minimum required thickness : 7.800 mm.

Minimum nozzle neck thickness 9.728 mm., is greater than required thickness 7.800 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	17.7,	Allowable	:	151.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	372.2 N./mm ²	Passed
Occasional	:	4.3,	Allowable	:	200.8 N./mm ²	Passed
Shear	:	9.4,	Allowable	:	105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[151.0/164.0, 1] \\ &= 0.921 \end{aligned}$$

Pad Material Factor [frp]:

$$\begin{aligned} &= \min[S_p/S, 1] \\ &= \min[164.0/164.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 9.735 - 3.0 \\ &= 6.735 \text{ mm.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2] \\ &= 0.9/6 \cdot 3506.0 [2 + (3506.0/(2 \cdot 878.0))^2] \\ &= 3148.218 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned} &= \sqrt{(Reff \cdot t) + W} \\ &= \sqrt{(3148.218 \cdot 17.0) + 38.65} \\ &= 269.993 \text{ mm.} \end{aligned}$$

Limit [LR2]:

$$\begin{aligned} &= \sqrt{((Reff + t) \cdot (t + t_e))} \\ &= \sqrt{((3148.218 + 17.0) \cdot (17.0 + 20.0))} \\ &= 342.218 \text{ mm.} \end{aligned}$$

Limit [LR3]:



$$\begin{aligned} &= 2 \cdot R_n \\ &= 2 \cdot 37.715 \\ &= 75.431 \text{ mm.} \end{aligned}$$

Final value of [LR]:

$$\begin{aligned} &= \min(LR1, LR2, LR3) \\ &= \min(269.993, 342.218, 75.431) \\ &= 75.431 \text{ mm.} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((17.0 + 20.0)/6.735)^{0.35}, 1) \\ &= \min(1.815, 1) \\ &= 1.000 \end{aligned}$$

Intermediate value [Cp]:

$$\begin{aligned} &= \exp((0.35 \cdot d_i - X_o) / (16t)) \\ &= \exp((0.35 \cdot 3506.0 - 44.45) / (16 \cdot 17.0)) \\ &= 77.322 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t) \\ &= \min(17.0 + 20.0 + 1.0 \cdot \sqrt{(37.715 \cdot 6.735)}, 400.0 + 17.0) \\ &= 52.937 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 17.0 + ((7.73 \cdot 1.0) / 75.431) \\ &= 27.248 \text{ mm.} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 17.0 \cdot 75.431 \\ &= 12.823 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 6.735 \cdot 52.937 \\ &= 3.565 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot Leg4^2 \\ &= 0.5 \cdot 20.0^2 \\ &= 2.000 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg41^2 - \text{Area cut by thickness limit} \\ &= 0.5 \cdot 20.0^2 - 0.083 \\ &= 1.917 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(38.65 \cdot 20.0, 75.431 \cdot 20.0) \\ &= 7.730 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5) \\ &= 12.823 + 0.921(3.565 + 0.0) + 1.917 + 2.0 + 0.0 + 1.0(7.73) \\ &= 27.753 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 6.735 / \ln[1 + 6.735/37.715] \\ &= 40.991 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 27.248 / \ln[1 + 27.248/3148.218] \end{aligned}$$



= 3161.835 mm.

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 164.0 \cdot 1.0 \\ &= 246.0 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 16.598 \cdot 40.991 \cdot 52.937 \\ &= 367.3 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n)/2 \\ &= 16.598 \cdot 3161.835 \cdot (75.431 + 6.735)/2 \\ &= 21986.1 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc})/2 \\ &= (16.598 \cdot 3161.835 \cdot 37.715)/2 \\ &= 10092.0 \text{ Kg} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (21986.1 + 10092.0 + 367.3)/16.598 \\ &= 1916.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff})) \\ &= 246.0 / (2 \cdot 1916.917 / 27.753 - 3161.835 / (2 \cdot 27.248)) \\ &= 30.7 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S / t / R_{xs} \\ &= 2 \cdot 164.0 / (17.0 / 3161.835) \\ &= 17.6 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(30.702, 17.634) \\ &= 17.634 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (367.278 + 21986.072 + 10092.012) / 27.753 \\ &= 114.648 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 16.598 \cdot 3161.835 / (2 \cdot 27.248) \\ &= 96.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 114.648 - 96.307, 96.307) \\ &= 133.0 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm ²
Local Primary Membrane Stress	PL	132.99	N./mm ²
Maximum Allowable Working Pressure	Pmax	17.63	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (37.715 + 6.735) / 37.715 \\ &= 1.179 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:



$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (37.715 + 6.735) \\ &= 69.822 \text{ mm.} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTP]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (37.715 + 6.735 + 38.65) \\ &= 130.533 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 14.142, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 14.142, \text{ must be } \geq \min(0.7 \cdot 6.735, 6.000) \text{ Passed} \\ t_{f1}, L42T &= 14.142, \text{ must be } \geq \min(0.6 \cdot 20.000, 0.6 \cdot 17.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(10092 \cdot 1.18, 1.5 \cdot 151.0(3.565 + 0.0), \pi/4 \cdot 16.6 \cdot 37.72^2 \cdot 1.18^2) \\ &= 262.653 \text{ Kg} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 262.7 \cdot 17.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164) \\ &= 120.678 \text{ Kg} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 262.7 \cdot 20.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164) \\ &= 141.975 \text{ Kg} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 120.678 / (69.822 \cdot (0.6 \cdot 17.0 + 0.49 \cdot 0.0)) \\ &= 1.662 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L41T)) \\ &= 141.975 / (69.822 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 14.142)) \\ &= 1.053 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (L_{TP} \cdot (0.49 \cdot L42T)) \\ &= 141.975 / (130.533 \cdot (0.49 \cdot 14.142)) \\ &= 1.539 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(1.662, 1.053, 1.539) \\ &= 1.7 \text{ must be less than or equal to } 164.0 \text{ N./mm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.058

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B



This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.058

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.391 • 1.0)/(9.735 - 3.0 - 0.0)
= 0.058

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.058

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.391 • 1.0)/(9.735 - 3.0 - 0.0)
= 0.058

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : 4 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 9 °C
Temperature reduction per Graph 3.12 : 5 °C
The computed stress ratio is : 0.882

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (14.989 • 1.0)/(20.0 - 3.0 - 0.0)
= 0.882

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.058

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.391 • 1.0)/(9.735 - 3.0 - 0.0)
= 0.058

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : 4 °C
Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -26 °C

The Drop for this Nozzle is : 0.3120 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 420.3119 mm.

Input Echo, WRC107/537 Item 1, Description: N43" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	6300.000	mm.
Vessel Thickness	Tv	20.000	mm.
Design Temperature	T1	65.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm2
Vessel Hot S.I. Allowable	Smh	164.00	N./mm2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	69.431	mm.
Nozzle Thickness	Tn	9.735	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	161.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	151.00	N./mm2
Thickness of Reinforcing Pad	Tpad	20.000	mm.
Diameter of Reinforcing Pad	Dpad	166.200	mm.
Design Internal Pressure	Dp	16.598	bars
Include Pressure Thrust		No	

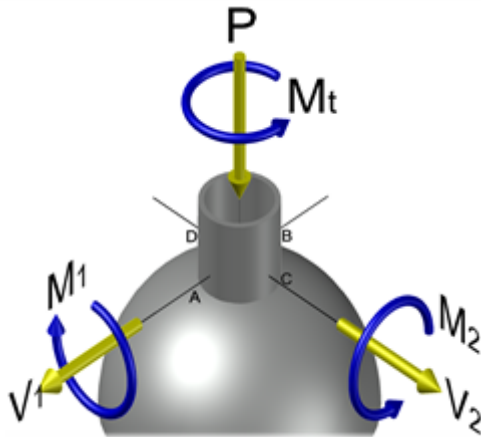
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	81.6	Kgf
Longitudinal Shear (SUS)	(V1) V1	122.4	Kgf
Circumferential Shear (SUS)	(Vc) V2	122.4	Kgf
Circumferential Moment (SUS)	(Mc) M1	24.5	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	36.7	Kg-m.
Torsional Moment (SUS)	Mt	44.9	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(3161.5 (20.0 - 3.0))}$
 $= 853.941 \text{ mm.}$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	81.6	Kgf
Circumferential Shear	(VC) V2	122.4	Kgf
Longitudinal Shear	(VL) V1	122.4	Kgf
Circumferential Moment	(MC) M1	24.5	Kg-m.
Longitudinal Moment	(ML) M2	36.7	Kg-m.
Torsional Moment	MT	44.9	Kg-m.

Dimensionless Param: U = 0.13 TAU = 6.10 RHO = 4.00 (5.49)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.10111	(A,B,C,D)
$M(x) / P$	SP 4	0.03922	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.42034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.18444	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.42034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.18444	(A,B,C,D)
$N(y) * T / P$	SP 4	0.33643	(A,B,C,D)
$M(y) / P$	SP 4	0.27446	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10306	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.41541	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10306	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.41541	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad. Bend.	P	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-0.6	0.6	0.6	-0.6
Rad. Memb.	ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-0.8	0.8	0.8	-0.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.4	0.6	1.0	-0.4	-1.0	0.4	0.6	-0.3



Tang. Memb. P	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Tang. Bend. P	-1.0	1.0	-1.0	1.0	-1.0	1.0	-1.0	1.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-4.3	4.3	4.3	-4.3
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.5	6.5	6.5	-6.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-7.8	7.2	5.4	-5.7	-5.6	5.1	3.2	-3.5
Shear VC	0.2	0.2	-0.2	-0.2	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.2	1.2	0.7	0.7	0.7	0.7	1.2	1.2
Str. Int.	8.0	7.4	5.6	5.8	5.7	5.2	3.7	4.0

Unitless Prm: U = 0.36 TAU = 0.00 (11.84) RHO = 0.00 (2.52)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.17200	(A,B,C,D)
M(x) / P	SR 2	0.11859	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16040	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.38504	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.16040	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.38504	(A,B,C,D)
N(y) * T / P	SR 2	0.05158	(A,B,C,D)
M(y) / P	SR 2	0.03556	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04931	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.11498	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04931	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.11498	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Rad. Bend. P		-2.0	2.0	-2.0	2.0	-2.0	2.0	-2.0	2.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-8.3	8.3	8.3	-8.3
Rad. Memb. ML		-0.9	-0.9	0.9	0.9	0.0	0.0	0.0	0.0
Rad. Bend. ML		-12.4	12.4	12.4	-12.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-15.7	13.0	10.8	-10.1	-11.3	9.2	6.4	-6.2
Tang. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Tang. Bend. P		-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-2.5	2.5	2.5	-2.5
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-3.7	3.7	3.7	-3.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.7	3.9	3.2	-3.0	-3.4	2.7	1.9	-1.8
Shear VC		0.3	0.3	-0.3	-0.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tot. Shear		0.9	0.9	0.3	0.3	0.3	0.3	0.9	0.9



Str. Int.	15.8	13.1	10.8	10.1	11.3	9.2	6.6	6.4
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WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad.	Pl (SUS)	-0.4	-0.4	0.3	0.3	-0.3	-0.3	0.2	0.2
Rad.	Q (SUS)	-1.0	1.0	0.7	-0.7	-0.7	0.7	0.4	-0.4
Long.	Pm (SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long.	Pl (SUS)	-0.3	-0.3	-0.1	-0.1	-0.2	-0.2	-0.1	-0.1
Long.	Q (SUS)	-7.5	7.5	5.6	-5.6	-5.3	5.3	3.4	-3.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.2	0.2	-0.2	-0.2	-0.2	-0.2	0.2	0.2
Shear	Q (SUS)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm	(SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl	(SUS)	71.0	71.0	71.5	71.5	71.1	71.1	71.4	71.4
Pm+Pl+Q	(Total)	70.0	78.6	76.7	70.8	70.3	76.3	74.8	71.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.51	246.00	Passed
Pm+Pl+Q (TOTAL)	78.55	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad.	Pl (SUS)	-1.3	-1.3	0.4	0.4	-1.1	-1.1	0.1	0.1
Rad.	Q (SUS)	-14.4	14.4	10.4	-10.4	-10.2	10.2	6.3	-6.3
Long.	Pm (SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long.	Pl (SUS)	-0.4	-0.4	0.1	0.1	-0.3	-0.3	0.0	0.0
Long.	Q (SUS)	-4.3	4.3	3.1	-3.1	-3.1	3.1	1.9	-1.9
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.3	0.3	-0.3	-0.3	-0.3	-0.3	0.3	0.3
Shear	Q (SUS)	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Pm	(SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl	(SUS)	70.8	70.8	71.7	71.7	70.9	70.9	71.5	71.5
Pm+Pl+Q	(Total)	66.5	84.3	82.0	68.2	67.8	80.3	77.7	69.5



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.69	246.00	Passed
Pm+Pl+Q (TOTAL)	84.26	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: N24"

From: 70

Pressure for Reinforcement Calculations	P	16.598	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm2
Parent Allowable Stress At Ambient	Sva	175.00	N./mm2
Inside Diameter of Elliptical Head	D	3500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm.
Head Internal Corrosion Allowance	c	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	151.00	N./mm2
Allowable Stress At Ambient	Sna	161.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm2
Pad Allowable Stress At Ambient	Spa	175.00	N./mm2
Diameter of Pad along vessel surface	Dp	204.3000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		45.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	3.734	in.
Actual Thickness Used in Calculation	0.383	in.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 100.831 \cdot (\exp(16.598/(151.0 \cdot 1.0)) - 1) + 3.0 + 0.0 \\ &= 0.5573 + 3.0 + 0.0 = 3.5573 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq} \\ &= 151.0 \cdot 1.0 \cdot \ln((2 \cdot 6.735 + 100.831)/100.831) - 0.0 \\ &= 189.315 - 0.0 = 189.3154 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 8.270 mm
Min. required shell thickness, internal p + c : 25.594 mm.

Nozzle neck minimum required thickness : 8.270 mm.

Minimum nozzle neck thickness 9.728 mm., is greater than required thickness 8.270 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	20.1,	Allowable	:	151.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	369.9 N./mm ²	Passed
Occasional	:	5.8,	Allowable	:	200.8 N./mm ²	Passed
Shear	:	9.6,	Allowable	:	105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[151.0/164.0, 1] \\ &= 0.921 \end{aligned}$$

Pad Material Factor [frp]:

$$\begin{aligned} &= \min[S_p/S, 1] \\ &= \min[164.0/164.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 9.735 - 3.0 \\ &= 6.735 \text{ mm.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2] \\ &= 0.9/6 \cdot 3506.0 [2 + (3506.0/(2 \cdot 878.0))^2] \\ &= 3148.218 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned} &= \sqrt{(Reff \cdot t) + W} \\ &= \sqrt{(3148.218 \cdot 17.0) + 45.0} \\ &= 276.343 \text{ mm.} \end{aligned}$$

Limit [LR2]:

$$\begin{aligned} &= \sqrt{((Reff + t) \cdot (t + t_e))} \\ &= \sqrt{((3148.218 + 17.0) \cdot (17.0 + 20.0))} \\ &= 342.218 \text{ mm.} \end{aligned}$$

Limit [LR3]:

$$= 2 \cdot R_n$$



$$= 2 \cdot 50.415 \\ = 100.831 \text{ mm.}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3) \\ = \min(276.343, 342.218, 100.831) \\ = 100.831 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + te)/tn)^{0.35}, 1) \\ = \min(((17.0 + 20.0)/6.735)^{0.35}, 1) \\ = \min(1.815, 1) \\ = 1.000$$

Angle for non-radial nozzles in Elliptical Heads [theta]:

$$= \text{atan}((h/R) \cdot (DR / (R^2 - DR^2))) \\ = \text{atan}((878.0/1753.0) \cdot (1000.0 / (1753.0^2 - 1000.0^2))) \\ = 0.335 \text{ radians}$$

Intermediate value [Cp]:

$$= \exp((0.35 \cdot di - X_o) / (16t)) \\ = \exp((0.35 \cdot 3506.0 - 1053.977) / (16 \cdot 17.0)) \\ = 1.890$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + te + Fp \cdot \sqrt{(Rn \cdot tn)}, Lpr1 + t) \\ = \min(17.0 + 20.0 + 1.0 \cdot \sqrt{(50.415 \cdot 6.735)}, 400.0 + 17.0) \\ = 55.426 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR) \\ = 17.0 + ((9.0 \cdot 1.0) / 100.831) \\ = 25.926 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \\ = 17.0 \cdot 100.831 \\ = 17.141 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH \\ = 6.735 \cdot 55.426 \\ = 3.733 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg42}^2 \\ = 0.5 \cdot 20.0^2 \\ = 2.000 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2 - \text{Area cut by thickness limit} \\ = 0.5 \cdot 20.0^2 - 0.012 \\ = 1.988 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot te, LR \cdot te) \\ = \min(45.0 \cdot 20.0, 100.831 \cdot 20.0) \\ = 9.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ = 17.141 + 0.921(3.733 + 0.0) + 1.988 + 2.0 + 0.0 + 1.0(9.0) \\ = 33.566 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn] \\ = 6.735 / \ln[1 + 6.735/50.415] \\ = 53.712 \text{ mm.}$$



Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 25.926 / \ln[1 + 25.926/3148.218] \\ &= 3161.157 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 164.0 \cdot 1.0 \\ &= 246.0 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 16.598 \cdot 53.712 \cdot 55.426 \\ &= 503.9 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n)/2 \\ &= 16.598 \cdot 3161.157 \cdot (100.831 + 6.735)/2 \\ &= 28776.5 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc})/2 \\ &= (16.598 \cdot 3161.157 \cdot 53.628)/2 \\ &= 14346.8 \text{ Kg} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (28776.5 + 14346.8 + 503.9)/16.598 \\ &= 2577.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff})) \\ &= 246.0 / (2 \cdot 2577.554 / 33.566 - 3161.157 / (2 \cdot 25.926)) \\ &= 26.6 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S / (t / R_{xs}) \\ &= 2 \cdot 164.0 / (17.0 / 3161.157) \\ &= 17.6 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(26.559, 17.638) \\ &= 17.638 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (503.894 + 28776.541 + 14346.783) / 33.566 \\ &= 127.465 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 16.598 \cdot 3161.157 / (2 \cdot 25.926) \\ &= 101.2 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 127.465 - 101.196, 101.196) \\ &= 153.7 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm2
Local Primary Membrane Stress	PL	153.73	N./mm2
Maximum Allowable Working Pressure	Pmax	17.64	bars

Strength of Nozzle Attachment Welds per 4.5.14



Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (53.628 + 6.735) / 53.628 \\ &= 1.126 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2} \\ &= \pi/2 \cdot \sqrt{((53.628 + 6.735)^2 + (50.415 + 6.735)^2) / 2} \\ &= 92.328 \text{ mm.} \end{aligned}$$

Weld Length of Pad to Shell Weld for non-radial Nozzles [LTp]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{((R_{nc} + t_n + W)^2 + (R_n + t_n + W)^2) / 2} \\ &= \pi/2 \cdot \sqrt{((53.628 + 6.735 + 45.0)^2 + (50.415 + 6.735 + 45.0)^2) / 2} \\ &= 162.999 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 14.142, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 14.142, \text{ must be } \geq \min(0.7 \cdot 6.735, 6.000) \text{ Passed} \\ t_f1, L42T &= 14.142, \text{ must be } \geq \min(0.6 \cdot 20.000, 0.6 \cdot 17.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwlds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(14347 \cdot 1.13, 1.5 \cdot 151.0(3.733 + 0.0), \pi/4 \cdot 16.6 \cdot 50.42^2 \cdot 1.13^2) \\ &= 428.075 \text{ Kg} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwlds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 428.1 \cdot 17.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164) \\ &= 196.683 \text{ Kg} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwlds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 428.1 \cdot 20.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164) \\ &= 231.392 \text{ Kg} \end{aligned}$$

Shear Stress [τ1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 196.683 / (92.328 \cdot (0.6 \cdot 17.0 + 0.49 \cdot 0.0)) \\ &= 2.048 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\ &= 231.392 / (92.328 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 14.142)) \\ &= 1.298 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ3]:

$$\begin{aligned} &= fwp / (L_{Tp} \cdot (0.49 \cdot L_{42T})) \\ &= 231.392 / (163.0 \cdot (0.49 \cdot 14.142)) \\ &= 2.009 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(2.048, 1.298, 2.009) \\ &= 2.0 \text{ must be less than or equal to } 164.0 \text{ N./mm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.078

The computed thickness ratio [sr]:



$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.522 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.078 \end{aligned}$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.078

The computed thickness ratio [sr]:
$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.522 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.078 \end{aligned}$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.078

The computed thickness ratio [sr]:
$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.522 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.078 \end{aligned}$$

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: 4 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: 9 °C
Temperature reduction per Graph 3.12	: 5 °C
The computed stress ratio is	: 0.882

The computed thickness ratio [sr]:
$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (14.989 \cdot 1.0) / (20.0 - 3.0 - 0.0) \\ &= 0.882 \end{aligned}$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.078

The computed thickness ratio [sr]:
$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.522 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.078 \end{aligned}$$

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: 4 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: 4 °C

Flange MDMT including temperature reduction per 3.11:



Flange MDMT with Temp reduction

-104 °C

The Drop for this Nozzle is : 19.6329 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 440.7045 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N24" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	6300.000	mm.
Vessel Thickness	Tv	20.000	mm.
Design Temperature	T1	65.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm2
Vessel Hot S.I. Allowable	Smh	164.00	N./mm2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	94.831	mm.
Nozzle Thickness	Tn	9.735	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	161.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	151.00	N./mm2
Thickness of Reinforcing Pad	Tpad	20.000	mm.
Diameter of Reinforcing Pad	Dpad	204.300	mm.
Design Internal Pressure	Dp	16.598	bars
Include Pressure Thrust		No	

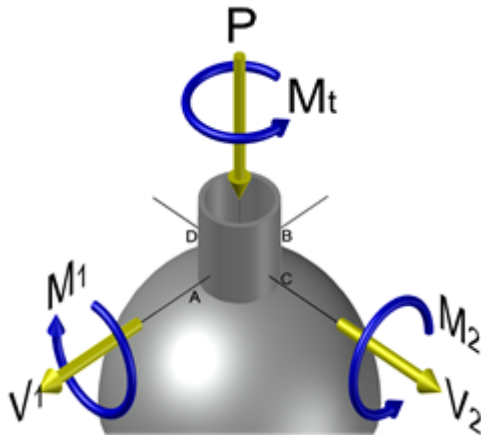
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	110.1	Kgf
Longitudinal Shear (SUS)	(V1) V1	163.2	Kgf
Circumferential Shear (SUS)	(Vc) V2	163.2	Kgf
Circumferential Moment (SUS)	(Mc) M1	48.9	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	65.3	Kg-m.
Torsional Moment (SUS)	Mt	81.6	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(3161.5 (20.0 - 3.0))}$$

$$= 879.341 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	110.1	Kgf
Circumferential Shear	(VC) V2	163.2	Kgf
Longitudinal Shear	(VL) V1	163.2	Kgf
Circumferential Moment	(MC) M1	48.9	Kg-m.
Longitudinal Moment	(ML) M2	65.3	Kg-m.
Torsional Moment	MT	81.6	Kg-m.

Dimensionless Param: U = 0.17 TAU = 7.99 RHO = 4.00 (5.49)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.08815	(A,B,C,D)
$M(x) / P$	SP 4	0.03691	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.32633	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.15470	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.32633	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.15470	(A,B,C,D)
$N(y) * T / P$	SP 4	0.34724	(A,B,C,D)
$M(y) / P$	SP 4	0.22680	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10227	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.07536	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10227	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.07536	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad. Bend.	P	-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-1.0	1.0	1.0	-1.0
Rad. Memb.	ML	-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-1.3	1.3	1.3	-1.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-2.0	0.9	1.5	-0.7	-1.5	0.7	1.0	-0.5

Tang. Memb. P	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
Tang. Bend. P	-1.1	1.1	-1.1	1.1	-1.1	1.1	-1.1	1.1
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-6.6	6.6	6.6	-6.6
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-8.8	8.8	8.8	-8.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-10.3	9.5	7.6	-7.9	-8.1	7.3	5.4	-5.7
Shear VC	0.2	0.2	-0.2	-0.2	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tot. Shear	1.3	1.3	0.8	0.8	0.8	0.8	1.3	1.3
Str. Int.	10.5	9.7	7.7	8.0	8.2	7.4	5.7	6.0

Unitless Prm: U = 0.44 TAU = 0.00 (14.67) RHO = 0.00 (2.52)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.14866	(A,B,C,D)
M(x) / P	SR 2	0.09572	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.15121	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.29323	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.15121	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.29323	(A,B,C,D)
N(y) * T / P	SR 2	0.04468	(A,B,C,D)
M(y) / P	SR 2	0.02892	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04592	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.08704	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04592	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.08704	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
Rad. Bend. P		-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Rad. Bend. MC		0.0	0.0	0.0	0.0	-12.6	12.6	12.6	-12.6
Rad. Memb. ML		-1.4	-1.4	1.4	1.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-16.8	16.8	16.8	-16.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-21.0	17.0	15.5	-13.8	-16.4	13.1	11.0	-9.9
Tang. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Tang. Bend. P		-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-3.7	3.7	3.7	-3.7
Tang. Memb. ML		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Tang. Bend. ML		-5.0	5.0	5.0	-5.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-6.2	5.0	4.6	-4.1	-4.9	3.9	3.3	-2.9
Shear VC		0.3	0.3	-0.3	-0.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Tot. Shear		1.0	1.0	0.4	0.4	0.4	0.4	1.0	1.0



Str. Int.	21.0	17.0	15.6	13.8	16.4	13.1	11.1	10.1
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WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad.	Pl (SUS)	-0.5	-0.5	0.4	0.4	-0.4	-0.4	0.3	0.3
Rad.	Q (SUS)	-1.4	1.4	1.1	-1.1	-1.1	1.1	0.8	-0.8
Long.	Pm (SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long.	Pl (SUS)	-0.4	-0.4	-0.1	-0.1	-0.4	-0.4	-0.2	-0.2
Long.	Q (SUS)	-9.9	9.9	7.7	-7.7	-7.7	7.7	5.5	-5.5
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.2	0.2	-0.2	-0.2	-0.2	-0.2	0.2	0.2
Shear	Q (SUS)	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Pm	(SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl	(SUS)	70.9	70.9	71.6	71.6	71.0	71.0	71.5	71.5
Pm+Pl+Q	(Total)	69.4	80.8	78.8	70.5	69.7	78.5	76.9	70.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.61	246.00	Passed
Pm+Pl+Q (TOTAL)	80.80	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad.	Pl (SUS)	-2.0	-2.0	0.9	0.9	-1.6	-1.6	0.5	0.5
Rad.	Q (SUS)	-19.0	19.0	14.7	-14.7	-14.8	14.8	10.5	-10.5
Long.	Pm (SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long.	Pl (SUS)	-0.6	-0.6	0.3	0.3	-0.5	-0.5	0.2	0.2
Long.	Q (SUS)	-5.6	5.6	4.3	-4.3	-4.4	4.4	3.1	-3.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.3	0.3	-0.3	-0.3	-0.3	-0.3	0.3	0.3
Shear	Q (SUS)	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Pm	(SUS)	71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl	(SUS)	70.6	70.6	72.1	72.1	70.7	70.7	71.8	71.8
Pm+Pl+Q	(Total)	65.0	88.2	86.7	67.1	66.3	84.3	82.3	68.4



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	72.14	246.00	Passed
Pm+Pl+Q (TOTAL)	88.18	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: N53"

From: 70

Pressure for Reinforcement Calculations	P	16.598	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm2
Parent Allowable Stress At Ambient	Sva	175.00	N./mm2
Inside Diameter of Elliptical Head	D	3500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm.
Head Internal Corrosion Allowance	c	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	151.00	N./mm2
Allowable Stress At Ambient	Sna	161.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm2
Pad Allowable Stress At Ambient	Spa	175.00	N./mm2
Diameter of Pad along vessel surface	Dp	166.2000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		38.6500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N53"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.734	in.
Actual Thickness Used in Calculation	0.383	in.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 75.431 \cdot (\exp(16.598/(151.0 \cdot 1.0)) - 1) + 3.0 + 0.0 \\ &= 0.4169 + 3.0 + 0.0 = 3.4169 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq} \\ &= 151.0 \cdot 1.0 \cdot \ln((2 \cdot 6.735 + 75.431)/75.431) - 0.0 \\ &= 248.071 - 0.0 = 248.0712 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.800 mm
Min. required shell thickness, internal p + c : 25.594 mm.

Nozzle neck minimum required thickness : 7.800 mm.

Minimum nozzle neck thickness 9.728 mm., is greater than required thickness 7.800 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	17.7	Allowable	:	151.0 N./mm ²	Passed
Expansion	:	0.0	Allowable	:	372.2 N./mm ²	Passed
Occasional	:	4.3	Allowable	:	200.8 N./mm ²	Passed
Shear	:	9.4	Allowable	:	105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[151.0/164.0, 1] \\ &= 0.921 \end{aligned}$$

Pad Material Factor [frp]:

$$\begin{aligned} &= \min[S_p/S, 1] \\ &= \min[164.0/164.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 9.735 - 3.0 \\ &= 6.735 \text{ mm.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2] \\ &= 0.9/6 \cdot 3506.0 [2 + (3506.0/(2 \cdot 878.0))^2] \\ &= 3148.218 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned} &= \sqrt{(Reff \cdot t) + W} \\ &= \sqrt{(3148.218 \cdot 17.0) + 38.65} \\ &= 269.993 \text{ mm.} \end{aligned}$$

Limit [LR2]:

$$\begin{aligned} &= \sqrt{((Reff + t) \cdot (t + t_e))} \\ &= \sqrt{((3148.218 + 17.0) \cdot (17.0 + 20.0))} \\ &= 342.218 \text{ mm.} \end{aligned}$$

Limit [LR3]:

$$= 2 \cdot R_n$$



$$= 2 \cdot 37.715 \\ = 75.431 \text{ mm.}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3) \\ = \min(269.993, 342.218, 75.431) \\ = 75.431 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + te)/tn)^{0.35}, 1) \\ = \min(((17.0 + 20.0)/6.735)^{0.35}, 1) \\ = \min(1.815, 1) \\ = 1.000$$

Angle for non-radial nozzles in Elliptical Heads [theta]:

$$= \text{atan}((h/R) \cdot (DR / (R^2 - DR^2))) \\ = \text{atan}((878.0/1753.0) \cdot (1000.0 / (1753.0^2 - 1000.0^2))) \\ = 0.335 \text{ radians}$$

Intermediate value [Cp]:

$$= \exp((0.35 \cdot di - Xo) / (16t)) \\ = \exp((0.35 \cdot 3506.0 - 1041.982) / (16 \cdot 17.0)) \\ = 1.975$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + te + Fp \cdot \sqrt{(Rn \cdot tn)}, Lpr1 + t) \\ = \min(17.0 + 20.0 + 1.0 \cdot \sqrt{(37.715 \cdot 6.735)}, 400.0 + 17.0) \\ = 52.937 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR) \\ = 17.0 + ((7.73 \cdot 1.0) / 75.431) \\ = 27.248 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \\ = 17.0 \cdot 75.431 \\ = 12.823 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH \\ = 6.735 \cdot 52.937 \\ = 3.565 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg42}^2 \\ = 0.5 \cdot 20.0^2 \\ = 2.000 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2 - \text{Area cut by thickness limit} \\ = 0.5 \cdot 20.0^2 - 0.083 \\ = 1.917 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot te, LR \cdot te) \\ = \min(38.65 \cdot 20.0, 75.431 \cdot 20.0) \\ = 7.730 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ = 12.823 + 0.921(3.565 + 0.0) + 1.917 + 2.0 + 0.0 + 1.0(7.73) \\ = 27.753 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn] \\ = 6.735 / \ln[1 + 6.735/37.715] \\ = 40.991 \text{ mm.}$$



Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 27.248 / \ln[1 + 27.248/3148.218] \\ &= 3161.835 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 164.0 \cdot 1.0 \\ &= 246.0 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 16.598 \cdot 40.991 \cdot 52.937 \\ &= 367.3 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n) / 2 \\ &= 16.598 \cdot 3161.835 \cdot (75.431 + 6.735) / 2 \\ &= 21986.1 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (16.598 \cdot 3161.835 \cdot 40.118) / 2 \\ &= 10735.0 \text{ Kg} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (21986.1 + 10735.0 + 367.3) / 16.598 \\ &= 1954.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff})) \\ &= 246.0 / (2 \cdot 1954.907 / 27.753 - 3161.835 / (2 \cdot 27.248)) \\ &= 29.7 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S / (t / R_{xs}) \\ &= 2 \cdot 164.0 / (17.0 / 3161.835) \\ &= 17.6 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(29.688, 17.634) \\ &= 17.634 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (367.278 + 21986.098 + 10735.03) / 27.753 \\ &= 116.921 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 16.598 \cdot 3161.835 / (2 \cdot 27.248) \\ &= 96.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 116.921 - 96.307, 96.307) \\ &= 137.5 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm2
Local Primary Membrane Stress	PL	137.53	N./mm2
Maximum Allowable Working Pressure	Pmax	17.63	bars

Strength of Nozzle Attachment Welds per 4.5.14



Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (40.118 + 6.735) / 40.118$$

$$= 1.168 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2}$$

$$= \pi/2 \cdot \sqrt{((40.118 + 6.735)^2 + (37.715 + 6.735)^2) / 2}$$

$$= 71.734 \text{ mm.}$$

Weld Length of Pad to Shell Weld for non-radial Nozzles [LTp]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + t_n + W)^2 + (R_n + t_n + W)^2) / 2}$$

$$= \pi/2 \cdot \sqrt{((40.118 + 6.735 + 38.65)^2 + (37.715 + 6.735 + 38.65)^2) / 2}$$

$$= 132.434 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 14.142, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 6.735, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 14.142, \text{ must be } \geq \min(0.6 \cdot 20.000, 0.6 \cdot 17.000) \text{ Passed}$$

Weld Load Value [fwlds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(10735 \cdot 1.17, 1.5 \cdot 151.0(3.565 + 0.0), \pi/4 \cdot 16.6 \cdot 37.72^2 \cdot 1.17^2)$$

$$= 257.908 \text{ Kgf}$$

Discontinuity Force [fws]:

$$= fwlds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 257.9 \cdot 17.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164)$$

$$= 118.498 \text{ Kgf}$$

Discontinuity Force [fwp]:

$$= fwlds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 257.9 \cdot 20.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164)$$

$$= 139.410 \text{ Kgf}$$

Shear Stress [τ1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 118.498 / (71.734 \cdot (0.6 \cdot 17.0 + 0.49 \cdot 0.0))$$

$$= 1.588 \text{ N./mm}^2$$

Shear Stress [τ2]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 139.41 / (71.734 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 14.142))$$

$$= 1.007 \text{ N./mm}^2$$

Shear Stress [τ3]:

$$= fwp / (L_{Tp} \cdot (0.49 \cdot L_{42T}))$$

$$= 139.41 / (132.434 \cdot (0.49 \cdot 14.142))$$

$$= 1.490 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(1.588, 1.007, 1.49)$$

$$= 1.6 \text{ must be less than or equal to } 164.0 \text{ N./mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.058

The computed thickness ratio [sr]:



$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.058

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.058

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: 4 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: 9 °C
Temperature reduction per Graph 3.12	: 5 °C
The computed stress ratio is	: 0.882

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (14.989 \cdot 1.0) / (20.0 - 3.0 - 0.0) \\ &= 0.882 \end{aligned}$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.058

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: 4 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: 4 °C

Flange MDMT including temperature reduction per 3.11:



Flange MDMT with Temp reduction

-104 °C

The Drop for this Nozzle is : 15.1646 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 436.2361 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N53" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	6300.000	mm.
Vessel Thickness	Tv	20.000	mm.
Design Temperature	T1	65.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm2
Vessel Hot S.I. Allowable	Smh	164.00	N./mm2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	69.431	mm.
Nozzle Thickness	Tn	9.735	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	161.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	151.00	N./mm2
Thickness of Reinforcing Pad	Tpad	20.000	mm.
Diameter of Reinforcing Pad	Dpad	166.200	mm.
Design Internal Pressure	Dp	16.598	bars
Include Pressure Thrust		No	

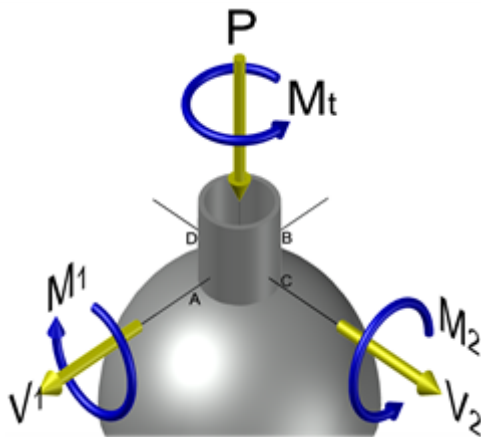
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	81.6	Kgf
Longitudinal Shear (SUS)	(V1) V1	122.4	Kgf
Circumferential Shear (SUS)	(Vc) V2	122.4	Kgf
Circumferential Moment (SUS)	(Mc) M1	24.5	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	36.7	Kg-m.
Torsional Moment (SUS)	Mt	44.9	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(3161.5 (20.0 - 3.0))}$
 $= 853.941 \text{ mm.}$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	81.6	Kgf
Circumferential Shear	(VC) V2	122.4	Kgf
Longitudinal Shear	(VL) V1	122.4	Kgf
Circumferential Moment	(MC) M1	24.5	Kg-m.
Longitudinal Moment	(ML) M2	36.7	Kg-m.
Torsional Moment	MT	44.9	Kg-m.

Dimensionless Param: U = 0.13 TAU = 6.10 RHO = 4.00 (5.49)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.10111	(A,B,C,D)
$M(x) / P$	SP 4	0.03922	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.42034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.18444	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.42034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.18444	(A,B,C,D)
$N(y) * T / P$	SP 4	0.33643	(A,B,C,D)
$M(y) / P$	SP 4	0.27446	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10306	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.41541	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10306	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.41541	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad. Bend.	P	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-0.6	0.6	0.6	-0.6
Rad. Memb.	ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-0.8	0.8	0.8	-0.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.4	0.6	1.0	-0.4	-1.0	0.4	0.6	-0.3



Tang. Memb. P	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Tang. Bend. P	-1.0	1.0	-1.0	1.0	-1.0	1.0	-1.0	1.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-4.3	4.3	4.3	-4.3
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.5	6.5	6.5	-6.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-7.8	7.2	5.4	-5.7	-5.6	5.1	3.2	-3.5
Shear VC	0.2	0.2	-0.2	-0.2	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.2	1.2	0.7	0.7	0.7	0.7	1.2	1.2
Str. Int.	8.0	7.4	5.6	5.8	5.7	5.2	3.7	4.0

Unitless Prm: U = 0.36 TAU = 0.00 (11.84) RHO = 0.00 (2.52)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.17200	(A,B,C,D)
M(x) / P	SR 2	0.11859	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16040	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.38504	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.16040	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.38504	(A,B,C,D)
N(y) * T / P	SR 2	0.05158	(A,B,C,D)
M(y) / P	SR 2	0.03556	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04931	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.11498	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04931	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.11498	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Rad. Bend. P		-2.0	2.0	-2.0	2.0	-2.0	2.0	-2.0	2.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-8.3	8.3	8.3	-8.3
Rad. Memb. ML		-0.9	-0.9	0.9	0.9	0.0	0.0	0.0	0.0
Rad. Bend. ML		-12.4	12.4	12.4	-12.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-15.7	13.0	10.8	-10.1	-11.3	9.2	6.4	-6.2
Tang. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Tang. Bend. P		-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-2.5	2.5	2.5	-2.5
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-3.7	3.7	3.7	-3.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.7	3.9	3.2	-3.0	-3.4	2.7	1.9	-1.8
Shear VC		0.3	0.3	-0.3	-0.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tot. Shear		0.9	0.9	0.3	0.3	0.3	0.3	0.9	0.9



Str. Int.	15.8	13.1	10.8	10.1	11.3	9.2	6.6	6.4
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WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad. Pl (SUS)		-0.4	-0.4	0.3	0.3	-0.3	-0.3	0.2	0.2
Rad. Q (SUS)		-1.0	1.0	0.7	-0.7	-0.7	0.7	0.4	-0.4
Long. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long. Pl (SUS)		-0.3	-0.3	-0.1	-0.1	-0.2	-0.2	-0.1	-0.1
Long. Q (SUS)		-7.5	7.5	5.6	-5.6	-5.3	5.3	3.4	-3.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.2	0.2	-0.2	-0.2	-0.2	-0.2	0.2	0.2
Shear Q (SUS)		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl (SUS)		71.0	71.0	71.5	71.5	71.1	71.1	71.4	71.4
Pm+Pl+Q (Total)		70.0	78.6	76.7	70.8	70.3	76.3	74.8	71.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.51	246.00	Passed
Pm+Pl+Q (TOTAL)	78.55	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad. Pl (SUS)		-1.3	-1.3	0.4	0.4	-1.1	-1.1	0.1	0.1
Rad. Q (SUS)		-14.4	14.4	10.4	-10.4	-10.2	10.2	6.3	-6.3
Long. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long. Pl (SUS)		-0.4	-0.4	0.1	0.1	-0.3	-0.3	0.0	0.0
Long. Q (SUS)		-4.3	4.3	3.1	-3.1	-3.1	3.1	1.9	-1.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.3	0.3	-0.3	-0.3	-0.3	-0.3	0.3	0.3
Shear Q (SUS)		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl (SUS)		70.8	70.8	71.7	71.7	70.9	70.9	71.5	71.5
Pm+Pl+Q (Total)		66.5	84.3	82.0	68.2	67.8	80.3	77.7	69.5



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.69	246.00	Passed
Pm+Pl+Q (TOTAL)	84.27	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: N63"

From: 70

Pressure for Reinforcement Calculations	P	16.598	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm2
Parent Allowable Stress At Ambient	Sva	175.00	N./mm2
Inside Diameter of Elliptical Head	D	3500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm.
Head Internal Corrosion Allowance	c	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	151.00	N./mm2
Allowable Stress At Ambient	Sna	161.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm2
Pad Allowable Stress At Ambient	Spa	175.00	N./mm2
Diameter of Pad along vessel surface	Dp	166.2000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		38.6500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N63"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.734	in.
Actual Thickness Used in Calculation	0.383	in.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 75.431 \cdot (\exp(16.598/(151.0 \cdot 1.0)) - 1) + 3.0 + 0.0 \\ &= 0.4169 + 3.0 + 0.0 = 3.4169 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq} \\ &= 151.0 \cdot 1.0 \cdot \ln((2 \cdot 6.735 + 75.431)/75.431) - 0.0 \\ &= 248.071 - 0.0 = 248.0712 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.800 mm
Min. required shell thickness, internal p + c : 25.594 mm.

Nozzle neck minimum required thickness : 7.800 mm.

Minimum nozzle neck thickness 9.728 mm., is greater than required thickness 7.800 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	17.7	Allowable	:	151.0 N./mm ²	Passed
Expansion	:	0.0	Allowable	:	372.2 N./mm ²	Passed
Occasional	:	4.3	Allowable	:	200.8 N./mm ²	Passed
Shear	:	9.4	Allowable	:	105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[151.0/164.0, 1] \\ &= 0.921 \end{aligned}$$

Pad Material Factor [frp]:

$$\begin{aligned} &= \min[S_p/S, 1] \\ &= \min[164.0/164.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 9.735 - 3.0 \\ &= 6.735 \text{ mm.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2] \\ &= 0.9/6 \cdot 3506.0 [2 + (3506.0/(2 \cdot 878.0))^2] \\ &= 3148.218 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned} &= \sqrt{(Reff \cdot t) + W} \\ &= \sqrt{(3148.218 \cdot 17.0) + 38.65} \\ &= 269.993 \text{ mm.} \end{aligned}$$

Limit [LR2]:

$$\begin{aligned} &= \sqrt{((Reff + t) \cdot (t + t_e))} \\ &= \sqrt{((3148.218 + 17.0) \cdot (17.0 + 20.0))} \\ &= 342.218 \text{ mm.} \end{aligned}$$

Limit [LR3]:

$$= 2 \cdot R_n$$



$$= 2 \cdot 37.715$$

$$= 75.431 \text{ mm.}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(269.993, 342.218, 75.431)$$

$$= 75.431 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + te)/tn)^{0.35}, 1)$$

$$= \min(((17.0 + 20.0)/6.735)^{0.35}, 1)$$

$$= \min(1.815, 1)$$

$$= 1.000$$

Angle for non-radial nozzles in Elliptical Heads [theta]:

$$= \text{atan}((h/R) \cdot (DR / (R^2 - DR^2)))$$

$$= \text{atan}((878.0/1753.0) \cdot (1000.0 / (1753.0^2 - 1000.0^2)))$$

$$= 0.335 \text{ radians}$$

Intermediate value [Cp]:

$$= \exp((0.35 \cdot di - Xo) / (16t))$$

$$= \exp((0.35 \cdot 3506.0 - 1041.982) / (16 \cdot 17.0))$$

$$= 1.975$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + te + Fp \cdot \sqrt{(Rn \cdot tn)}, Lpr1 + t)$$

$$= \min(17.0 + 20.0 + 1.0 \cdot \sqrt{(37.715 \cdot 6.735)}, 400.0 + 17.0)$$

$$= 52.937 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 17.0 + ((7.73 \cdot 1.0) / 75.431)$$

$$= 27.248 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 17.0 \cdot 75.431$$

$$= 12.823 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 6.735 \cdot 52.937$$

$$= 3.565 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg42}^2$$

$$= 0.5 \cdot 20.0^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2 - \text{Area cut by thickness limit}$$

$$= 0.5 \cdot 20.0^2 - 0.083$$

$$= 1.917 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot te, LR \cdot te)$$

$$= \min(38.65 \cdot 20.0, 75.431 \cdot 20.0)$$

$$= 7.730 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 12.823 + 0.921(3.565 + 0.0) + 1.917 + 2.0 + 0.0 + 1.0(7.73)$$

$$= 27.753 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 6.735 / \ln[1 + 6.735/37.715]$$

$$= 40.991 \text{ mm.}$$



Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 27.248 / \ln[1 + 27.248/3148.218] \\ &= 3161.835 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 164.0 \cdot 1.0 \\ &= 246.0 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 16.598 \cdot 40.991 \cdot 52.937 \\ &= 367.3 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n) / 2 \\ &= 16.598 \cdot 3161.835 \cdot (75.431 + 6.735) / 2 \\ &= 21986.1 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (16.598 \cdot 3161.835 \cdot 40.118) / 2 \\ &= 10735.0 \text{ Kg} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (21986.1 + 10735.0 + 367.3) / 16.598 \\ &= 1954.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff})) \\ &= 246.0 / (2 \cdot 1954.907 / 27.753 - 3161.835 / (2 \cdot 27.248)) \\ &= 29.7 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S / (t / R_{xs}) \\ &= 2 \cdot 164.0 / (17.0 / 3161.835) \\ &= 17.6 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(29.688, 17.634) \\ &= 17.634 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (367.278 + 21986.098 + 10735.03) / 27.753 \\ &= 116.921 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 16.598 \cdot 3161.835 / (2 \cdot 27.248) \\ &= 96.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 116.921 - 96.307, 96.307) \\ &= 137.5 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm2
Local Primary Membrane Stress	PL	137.53	N./mm2
Maximum Allowable Working Pressure	Pmax	17.63	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (40.118 + 6.735) / 40.118$$

$$= 1.168 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2}$$

$$= \pi/2 \cdot \sqrt{((40.118 + 6.735)^2 + (37.715 + 6.735)^2) / 2}$$

$$= 71.734 \text{ mm.}$$

Weld Length of Pad to Shell Weld for non-radial Nozzles [LTp]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + t_n + W)^2 + (R_n + t_n + W)^2) / 2}$$

$$= \pi/2 \cdot \sqrt{((40.118 + 6.735 + 38.65)^2 + (37.715 + 6.735 + 38.65)^2) / 2}$$

$$= 132.434 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 14.142, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 6.735, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 14.142, \text{ must be } \geq \min(0.6 \cdot 20.000, 0.6 \cdot 17.000) \text{ Passed}$$

Weld Load Value [fwlds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(10735 \cdot 1.17, 1.5 \cdot 151.0(3.565 + 0.0), \pi/4 \cdot 16.6 \cdot 37.72^2 \cdot 1.17^2)$$

$$= 257.908 \text{ Kgf}$$

Discontinuity Force [fws]:

$$= fwlds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 257.9 \cdot 17.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164)$$

$$= 118.498 \text{ Kgf}$$

Discontinuity Force [fwp]:

$$= fwlds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 257.9 \cdot 20.0 \cdot 164 / (17.0 \cdot 164 + 20.0 \cdot 164)$$

$$= 139.410 \text{ Kgf}$$

Shear Stress [τ1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 118.498 / (71.734 \cdot (0.6 \cdot 17.0 + 0.49 \cdot 0.0))$$

$$= 1.588 \text{ N./mm}^2$$

Shear Stress [τ2]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L41T))$$

$$= 139.41 / (71.734 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 14.142))$$

$$= 1.007 \text{ N./mm}^2$$

Shear Stress [τ3]:

$$= fwp / (L_{Tp} \cdot (0.49 \cdot L42T))$$

$$= 139.41 / (132.434 \cdot (0.49 \cdot 14.142))$$

$$= 1.490 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(1.588, 1.007, 1.49)$$

$$= 1.6 \text{ must be less than or equal to } 164.0 \text{ N./mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.058

The computed thickness ratio [sr]:



$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.058

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.058

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: 4 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: 9 °C
Temperature reduction per Graph 3.12	: 5 °C
The computed stress ratio is	: 0.882

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (14.989 \cdot 1.0) / (20.0 - 3.0 - 0.0) \\ &= 0.882 \end{aligned}$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.058

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.391 \cdot 1.0) / (9.735 - 3.0 - 0.0) \\ &= 0.058 \end{aligned}$$

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: 4 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: 4 °C

Flange MDMT including temperature reduction per 3.11:



Flange MDMT with Temp reduction

-104 °C

The Drop for this Nozzle is : 15.1646 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 436.2361 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N63" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	6300.000	mm.
Vessel Thickness	Tv	20.000	mm.
Design Temperature	T1	65.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm2
Vessel Hot S.I. Allowable	Smh	164.00	N./mm2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	69.431	mm.
Nozzle Thickness	Tn	9.735	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	161.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	151.00	N./mm2
Thickness of Reinforcing Pad	Tpad	20.000	mm.
Diameter of Reinforcing Pad	Dpad	166.200	mm.
Design Internal Pressure	Dp	16.598	bars
Include Pressure Thrust		No	

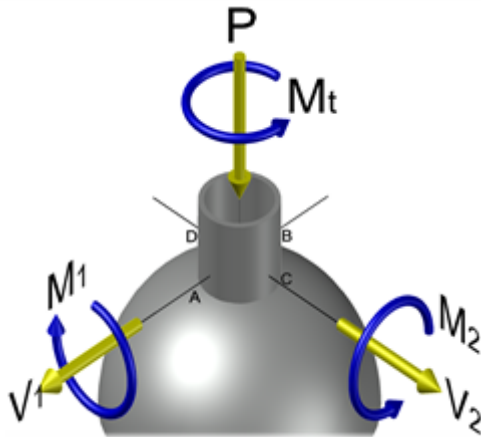
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	81.6	Kgf
Longitudinal Shear (SUS)	(V1) V1	122.4	Kgf
Circumferential Shear (SUS)	(Vc) V2	122.4	Kgf
Circumferential Moment (SUS)	(Mc) M1	24.5	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	36.7	Kg-m.
Torsional Moment (SUS)	Mt	44.9	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(3161.5 (20.0 - 3.0))}$
 $= 853.941 \text{ mm.}$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	81.6	Kgf
Circumferential Shear	(VC) V2	122.4	Kgf
Longitudinal Shear	(VL) V1	122.4	Kgf
Circumferential Moment	(MC) M1	24.5	Kg-m.
Longitudinal Moment	(ML) M2	36.7	Kg-m.
Torsional Moment	MT	44.9	Kg-m.

Dimensionless Param: U = 0.13 TAU = 6.10 RHO = 4.00 (5.49)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.10111	(A,B,C,D)
$M(x) / P$	SP 4	0.03922	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.42034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.18444	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.42034	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.18444	(A,B,C,D)
$N(y) * T / P$	SP 4	0.33643	(A,B,C,D)
$M(y) / P$	SP 4	0.27446	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10306	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.41541	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10306	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.41541	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad. Bend.	P	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-0.6	0.6	0.6	-0.6
Rad. Memb.	ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-0.8	0.8	0.8	-0.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.4	0.6	1.0	-0.4	-1.0	0.4	0.6	-0.3



Tang. Memb. P	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Tang. Bend. P	-1.0	1.0	-1.0	1.0	-1.0	1.0	-1.0	1.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-4.3	4.3	4.3	-4.3
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.5	6.5	6.5	-6.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-7.8	7.2	5.4	-5.7	-5.6	5.1	3.2	-3.5
Shear VC	0.2	0.2	-0.2	-0.2	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.2	1.2	0.7	0.7	0.7	0.7	1.2	1.2
Str. Int.	8.0	7.4	5.6	5.8	5.7	5.2	3.7	4.0

Unitless Prm: U = 0.36 TAU = 0.00 (11.84) RHO = 0.00 (2.52)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.17200	(A,B,C,D)
M(x) / P	SR 2	0.11859	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16040	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.38504	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.16040	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.38504	(A,B,C,D)
N(y) * T / P	SR 2	0.05158	(A,B,C,D)
M(y) / P	SR 2	0.03556	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04931	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.11498	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04931	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.11498	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Rad. Bend. P		-2.0	2.0	-2.0	2.0	-2.0	2.0	-2.0	2.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-8.3	8.3	8.3	-8.3
Rad. Memb. ML		-0.9	-0.9	0.9	0.9	0.0	0.0	0.0	0.0
Rad. Bend. ML		-12.4	12.4	12.4	-12.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-15.7	13.0	10.8	-10.1	-11.3	9.2	6.4	-6.2
Tang. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Tang. Bend. P		-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-2.5	2.5	2.5	-2.5
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-3.7	3.7	3.7	-3.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.7	3.9	3.2	-3.0	-3.4	2.7	1.9	-1.8
Shear VC		0.3	0.3	-0.3	-0.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tot. Shear		0.9	0.9	0.3	0.3	0.3	0.3	0.9	0.9



Str. Int.	15.8	13.1	10.8	10.1	11.3	9.2	6.6	6.4
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WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad. Pl (SUS)		-0.4	-0.4	0.3	0.3	-0.3	-0.3	0.2	0.2
Rad. Q (SUS)		-1.0	1.0	0.7	-0.7	-0.7	0.7	0.4	-0.4
Long. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long. Pl (SUS)		-0.3	-0.3	-0.1	-0.1	-0.2	-0.2	-0.1	-0.1
Long. Q (SUS)		-7.5	7.5	5.6	-5.6	-5.3	5.3	3.4	-3.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.2	0.2	-0.2	-0.2	-0.2	-0.2	0.2	0.2
Shear Q (SUS)		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl (SUS)		71.0	71.0	71.5	71.5	71.1	71.1	71.4	71.4
Pm+Pl+Q (Total)		70.0	78.6	76.7	70.8	70.3	76.3	74.8	71.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.51	246.00	Passed
Pm+Pl+Q (TOTAL)	78.55	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Rad. Pl (SUS)		-1.3	-1.3	0.4	0.4	-1.1	-1.1	0.1	0.1
Rad. Q (SUS)		-14.4	14.4	10.4	-10.4	-10.2	10.2	6.3	-6.3
Long. Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Long. Pl (SUS)		-0.4	-0.4	0.1	0.1	-0.3	-0.3	0.0	0.0
Long. Q (SUS)		-4.3	4.3	3.1	-3.1	-3.1	3.1	1.9	-1.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.3	0.3	-0.3	-0.3	-0.3	-0.3	0.3	0.3
Shear Q (SUS)		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Pm (SUS)		71.1	71.1	71.1	71.1	71.1	71.1	71.1	71.1
Pm+Pl (SUS)		70.8	70.8	71.7	71.7	70.9	70.9	71.5	71.5
Pm+Pl+Q (Total)		66.5	84.3	82.0	68.2	67.8	80.3	77.7	69.5



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	71.14	164.00	Passed
Pm+Pl (SUS)	71.69	246.00	Passed
Pm+Pl+Q (TOTAL)	84.27	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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